Energy and AI: Expanded Research Excerpt

International Energy Agency | 2025

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Included original page ranges:

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13-17: Executive summary

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Selection focuses on data-center energy, cooling, water and supply options.

Some boundary pages also contain the start of an adjacent section.

Reference sections are retained, including entries for omitted material.

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The full source remains available as article_10.pdf.

Energy and AI



International Energy Agency

World Energy Outlook Special Report

INTERNATIONAL ENERGY AGENCY

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The transformative potential of AI depends on energy

There has been a step change in the capabilities of artificial intelligence (AI), driven by falling computation costs, a surge in data availability and technical breakthroughs. AI is the science of making machines capable of learning to perform tasks that traditionally required human intelligence. AI is emerging as a general-purpose technology, much like electricity. Today, it can generate text and videos, accelerate scientific discovery in fields like medicine or materials science, make manufacturing robots smarter and more productive, drive commercial taxis in complex city landscapes, and detect threats to critical infrastructure.

In the past few years, AI has gone from an academic pursuit to an industry with trillions of dollars of market capitalisation and venture capital at stake. The market capitalisation of AI-related firms in the S&P 500 has grown by around USD 12 trillion since 2022. While there are several uncertainties about its uptake and impact, AI's rapid development and huge potential have made it central to corporate strategies, economic policies and geopolitics.

However, there is no AI without energy; at the same time, AI has the potential to transform the energy sector. Affordable, reliable and sustainable electricity supply will be a crucial determinant of AI development, and countries that can deliver the energy needed at speed and scale will be best placed to benefit. Training and deploying AI models takes place in large and power-hungry data centres. A typical AI-focused data centre consumes as much electricity as 100 000 households, but the largest ones under construction today will consume 20 times as much.

Policy makers and markets have lacked the tools to assess implications

The energy sector is therefore at the heart of one of the most important technological revolutions today. However, there is still a lack of understanding of the stakes and implications of this deepening connection between energy and AI. Consistent with its strong track record of identifying and exploring emerging issues in the energy sector, this new International Energy Agency (IEA) special report seeks to fill this gap with the most comprehensive, data-driven analysis on the topic to date. Based on a new global model and comprehensive dataset of data centre electricity demand, its analysis was also enriched by an in-depth process of consultation with policy makers, the tech sector, the energy industry and other experts.

Data centres account for a small share of global electricity consumption today, but their local impacts are far more pronounced

Global investment in data centres has nearly doubled since 2022 and amounted to half a trillion dollars in 2024. This investment boom has led to growing concerns about skyrocketing electricity demand.

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Data centres accounted for around 1.5% of the world's electricity consumption in 2024, or 415 terawatt-hours (TWh). The United States accounted for the largest share of global data centre electricity consumption in 2024 (45%), followed by China (25%) and Europe (15%). Globally, data centre electricity consumption has grown by around 12% per year since 2017, more than four times faster than the rate of total electricity consumption. AI-focused data centres can draw as much electricity as power-intensive factories such as aluminium smelters, but they are much more geographically concentrated. Nearly half of data centre capacity in the United States is in five regional clusters. The sector accounts for substantial shares of electricity consumption in local markets.

Electricity demand for data centres more than doubles by 2030

Data centre electricity consumption is set to more than double to around 945 TWh by 2030. This is slightly more than Japan's total electricity consumption today. AI is the most important driver of this growth, alongside growing demand for other digital services. The United States accounts for by far the largest share of this projected increase, followed by China. In the United States, data centres account for nearly half of electricity demand growth between now and 2030. By the end of the decade, the country is set to consume more electricity for data centres than for the production of aluminium, steel, cement, chemicals and all other energy-intensive goods combined. Uncertainties widen further after 2030, but our Base Case sees global data centre electricity consumption rising to around 1 200 TWh by 2035.

A diverse range of sources will be needed to meet demand

Renewables and natural gas take the lead in meeting data centre electricity demand, but a range of sources are poised to contribute. Half of the global growth in data centre demand is met by renewables, supported by storage and the broader electricity grid. Renewables generation is projected to grow by over 450 TWh to meet data centre demand to 2035, building on short lead times, economic competitiveness and the procurement strategies of tech companies. Dispatchable sources, led by natural gas, also have a crucial role to play, with the tech sector helping to bring forward new nuclear and geothermal technologies as well. Natural gas expands by 175 TWh to meet growing data centre demand, notably in the United States. Nuclear contributes about the same amount of additional generation to meet data centre demand, notably in China, Japan and the United States. The first small modular reactors come online around 2030.

Data centres are one of several drivers of accelerated electricity demand growth in the Age of Electricity

Data centres account for around one-tenth of global electricity demand growth to 2030, less than the share from industrial motors, air conditioning in homes and offices, or electric vehicles. However, the significance of data centres in driving electricity demand differs by country. Emerging and developing economies are already experiencing rapid electricity

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demand growth. In these countries, data centres account for around 5% of the increase in electricity demand to 2030. Advanced economies, on the other hand, have seen several decades of essentially stagnant electricity demand. In this group of countries, data centres account for more than 20% of demand growth to 2030, presenting a wake-up call on the need to put the electricity sector on a growth footing again.

Smarter is faster when it comes to integrating data centres in electricity grids

Electricity grids are already under strain in many places: we estimate that unless these risks are addressed, around 20% of planned data centre projects could be at risk of delays. Grid connection queues for both supply and consumption projects, including data centres, are long and complex. Building new transmission lines can take four to eight years in advanced economies and wait times for critical grid components such as transformers and cables have doubled in the past three years. Generation equipment is also in high demand. Turbine deliveries for new gas-fired power plants now face lead times of several years, potentially delaying their commissioning beyond 2030. If the electricity sector does not step up, there is a risk that meeting data centre load growth could entail trade-offs with other goals such as electrification, manufacturing growth or affordability.

Key options to mitigate these risks include locating new data centres in areas of high power and grid availability, and operating either data centre servers or their onsite power generation and storage assets more flexibly. These strategies are still underexplored. An AI-focused data centre is 10 times more capital-intensive than an aluminium smelter, which means curtailing its operations to provide flexibility to the grid is very costly. But many data centres operate with a buffer of spare server capacity. Regulators could explore measures to incentivise data centre operators to use spare server capacity or their backup power generation or storage assets more flexibly. Grid operators could also examine incentives to locate data centres in areas where grids are less constrained. We find that 50% of data centres under development in the United States are in pre-existing large clusters, potentially raising risks of local bottlenecks.

There are large uncertainties in the outlook for AI-related electricity demand

There are uncertainties in how quickly AI will be adopted, how capable and productive it will become, how fast efficiency improvements will occur, and whether bottlenecks in the energy sector can be resolved. These uncertainties are explored in sensitivity cases. A Lift-Off Case assumes higher rates of AI uptake and proactive action to reduce energy sector bottlenecks. A Headwinds Case incorporates bottlenecks – including macroeconomic headwinds – in the uptake of AI and the buildout of energy infrastructure to power it. Our High Efficiency Case highlights the potential for even stronger gains in the efficiency of AI-related hardware and AI models. In this case, electricity demand from data centres is 20% lower in 2035 than in the Base Case. By 2035, the range of data centre electricity demand across our cases spans from 700 to 1 700 TWh. The increase in gas-fired power to meet data

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centre demand in our Lift-Off Case is four times higher than in our Headwinds Case. Growth in nuclear output to meet data centre demand varies even more.

AI could unlock major efficiency and operational gains for the energy sector

AI is already being deployed by energy companies to transform and optimise energy and mineral supply, electricity generation and transmission, and energy consumption. There are numerous objectives in play, including reducing costs, enhancing supply, extending asset lifetimes, reducing downtime and lowering emissions.

The oil and gas industry has been an early adopter of AI, using it to optimise exploration, production, maintenance and safety. In exploration and development, AI can make the evaluation of resources more reliable and reduce predrilling uncertainty. In operations, it is being used to optimise and automate production processes, detect leaks, predict maintenance needs, and support efforts to reduce methane emissions.

AI can help to balance electricity networks that are growing more complex, decentralised and digitalised. AI can improve the forecasting and integration of variable renewable energy generation, reducing curtailment and emissions. AI-based fault detection can help rapidly identify and precisely pinpoint grid faults, reducing outage durations by 30-50%. Remote sensors and AI-based management can increase the capacity of transmission lines. Up to 175 gigawatts (GW) of transmission capacity could be unlocked if these tools are applied, without any new lines being built. This is more than the increase in the data centre power load to 2030 in the Base Case.

The industry of the future will be increasingly digitalised and automated; countries and companies that take the lead in integrating AI into manufacturing will jump ahead. AI applications can accelerate product development, lower costs and increase quality. Widespread adoption of existing AI applications to optimise processes in industry can lead to energy savings equivalent to more than the total energy consumption of Mexico today. European companies have over half of the market share for industrial automation solutions, which are the critical enabler for industrial AI deployment.

AI applications in transport can improve efficiency and save costs, but they could also increase demand for personal mobility. AI applications are being used to manage traffic, optimise routes, predict maintenance needs and develop autonomous vehicles. The widespread adoption of AI applications across the transport sector could lead to energy savings equivalent to the energy used by 120 million cars. While autonomous vehicles operate more efficiently than conventional ones, they might also attract people away from public transport as costs fall and availability increases, leading to rebound effects.

In buildings, there is significant potential for AI-led optimisations to make heating and cooling systems more efficient and electricity use in buildings more flexible. Barriers to realising this potential include fragmented ownership of buildings, lack of digitalisation and inadequate incentives. If scaled up, existing AI-led interventions could lead to global

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electricity savings of around 300 TWh, equivalent to annual electricity generation today for Australia and New Zealand combined.

Accelerated innovation could be one of the most significant longer-term impacts of AI on the energy sector

AI is emerging as a powerful tool for scientific discovery, helping researchers to find, test and commercialise innovations faster. In biomedicine, for example, AI led to a 45 000-fold acceleration in the mapping of protein structures – critical for designing new drugs. Innovation lead times for new energy technologies often span decades. Reducing this period will be key to achieving energy sector goals such as sustainability and competitiveness. Yet only 2% of the equity raised by energy start-ups has gone to companies with an AI-related value proposition.

Energy innovation challenges are characterised by the kinds of problems AI is good at solving. For example, only 0.01% of next-generation solar PV materials have been experimentally produced, leaving a huge set of possible materials still to be explored. AI could allow scientists to dramatically accelerate the process of finding and testing promising materials, battery chemistries and carbon capture molecules. Policy will be required to support AI-led invention and also accelerate commercialisation, which is often a bigger impediment to new products than the discovery phase.

The energy sector is not yet making the most of AI

Energy is amongst the most complex and critical sectors in the world today, yet it can and should do more to seize the potential benefits of harnessing AI. The energy sector faces barriers to realising the widespread adoption of AI, including missing or inadequate access to data and digital infrastructure and skills, as well as persistent digital and physical security concerns, which often trump potential efficiency gains. The prevalence of AI-related skills is much lower in the energy sector compared with other sectors. Policy and regulatory changes will be needed to enable the energy sector to seize the benefits of AI.

AI could sharpen some energy security concerns and help address others

The supply chains for the components going into data centres are complex and globalised. For example, gallium is an increasingly critical metal used in cutting-edge computer chips and power electronics, offering significant efficiency benefits compared with traditional silicon-based semiconductor designs. China currently accounts for around 99% of global refined gallium supply. Our estimates indicate that in 2030, demand for gallium for data centres could reach over 10% of today's supply.

AI compounds some energy security risks, but it also offers solutions in both the cyber and physical domains. As AI capabilities increase, so does the capacity for them to be used and misused by various actors. Cyberattacks on energy utilities have tripled in the past four years and have become more sophisticated because of AI. At the same time, AI is becoming a

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Energy for AI

The evolution of energy demand and how to meet it

SUMMARY

- Artificial intelligence (AI) model training and deployment occur mainly in data centres. In total, electricity consumption from data centres is estimated to amount to around 415 terawatt hours (TWh), or about 1.5% of global electricity consumption in 2024. It has grown at 12% per year over the last five years.
- Our Base Case projections for data centre electricity consumption are grounded in the latest industry expectations for server shipments. Three sensitivity cases (Lift-Off, High Efficiency and Headwinds) capture uncertainties in efficiency improvements in hardware and software, AI uptake and energy sector bottlenecks.
- In the Base Case, electricity consumption from data centres rises to around 945 TWh by 2030, more than doubling from the 2024 level. The United States sees by far the largest absolute growth, followed by China and Europe. Data centres still account for less than 10% of the growth in global electricity consumption to 2030.
- The Lift-Off Case assumes stronger AI uptake and limited local constraints on data centre buildout. In this case, consumption reaches over 1 260 TWh by 2030. The High Efficiency Case is driven by energy savings in both software and hardware; consumption reaches around 800 TWh by 2030. In the Headwinds Case, it reaches around 670 TWh. By 2035, the spread of uncertainty widens further, spanning 700 TWh to 1 720 TWh across the four cases.
- Natural gas generation to meet data centre demand increases by around 175 TWh from today's level to 2035 in the Base Case, mostly concentrated in the United States. In the Lift-Off Case, it grows by 290 TWh. Renewables provide the largest contribution to meet data centre demand, increasing by 450 TWh to 2035 in the Base Case. This reflects their broad availability, short development times, economic competitiveness and technology sector procurement strategies. Nuclear power also contributes.
- Grid congestion and connection queues are growing in many regions, and supply chains for key components like transformers and gas turbines are stretched. In our analysis and modelling of these factors, we estimate that around 20% of the projected data centre additions by 2030 in our Base Case could be at risk of delay.
- Avoiding this risk will require a range of actions from both the energy and technology sectors. Permitting times for new projects need to be cut. Grid operators should streamline the confusing tangle of data centre connection applications. The technology sector should maximise the buildout of data centres in areas of high power and grid availability and explore strategies to incentivise their operational flexibility. Better management of the growing data centre load could be facilitated by better data on both grid constraints and the data centre demand outlook.

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2.1 Introduction

Investment in new data centres has surged, increasing by nearly 70% in the last two years at the global level. One of the main drivers of this investment has been the rise of artificial intelligence (AI), alongside the deepening digitalisation of the global economy. The rapid increase in data centre investment is raising concerns about the ability of electricity systems to meet growing demand in a timely, secure and sustainable way.

Data centres – at least at the scale seen today – are relatively new actors in the energy system at the global level, and data collection and reporting on their electricity consumption remain limited. There is therefore substantial uncertainty about both their current and future consumption. Moreover, AI models are highly heterogeneous, and data on their uptake and electricity intensity are limited (see Chapter 1). As a result, it is challenging to analyse the link between AI demand and data centre electricity consumption.

On the electricity supply side of the equation, the sector is facing several challenges. Electricity demand is already growing strongly in emerging market and developing economies, driven especially by economic growth, industrialisation, increased adoption of appliances, and surging needs for cooling. Advanced economies are also returning to growth in electricity demand after two decades of stagnation. However, the electricity sector faces several bottlenecks, including permitting times and tangled supply chains.

This chapter explores these issues across the following sections:

- Section 2.2 sets the scene by describing the determinants of data centre electricity consumption and how much electricity data centres consume today.
- Section 2.3 presents new International Energy Agency (IEA) modelling on the outlook for electricity demand from data centres.
- Section 2.4 places data centres within the broader context of the information and communication technology (ICT) sector and discusses how the uptake of AI may influence the energy consumption of the ICT sector beyond data centres.
- Section 2.5 examines electricity supply scenarios to meet the demand growth from data centres.
- Section 2.6 discusses how data centres interact with grids and what can be done to avert the risk of project delays due to electricity sector constraints.

2.1.1 Case design

The uncertainty surrounding future data centre electricity demand requires a scenario-based approach to explore alternative pathways and provide perspectives on timelines relevant to energy sector decision making. While the technology sector moves quickly and a data centre can be operational in two to three years, the broader energy system requires longer lead times to schedule and build infrastructure, which often requires extensive planning, long build times and high upfront investment. At the same time, information on the project

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pipeline for new data centres extends only a few years into the future. Industry forecasts for key variables – such as chip and server production and shipments – are likewise short term (three to five years), reflecting the rapid pace of innovation and inherent uncertainties surrounding key drivers of data centre demand, notably AI uptake.

For these reasons, we present our cases across two timeframes. Section 2.3.1 focuses on projections to 2030 and the results of our Base Case (see below). Section 2.3.2 takes a slightly longer-term, exploratory approach, presenting results to 2035 and for a wider range of cases. The underlying assumptions of these four cases are briefly described as follows, with more details provided in sections 2.3.1 and 2.3.2:

- The **Base Case** explores the trajectory of electricity consumption in data centres under current regulatory conditions and industry projections. The key driver in this case in the near term is industry projections for server shipments to 2028 and a continuation of this trend after 2028. Efficiency improvements are expected to continue playing a pivotal role in limiting strong growth in energy consumption, despite increasing demand for digital services.
- The **Lift-Off Case** assumes stronger growth in AI adoption than in the Base Case. A more resilient supply chain and greater flexibility in data centre location, powering and operations enable faster data centre deployment.
- The **High Efficiency Case** shares similar constraints and drivers with the Base Case but assumes stronger progress on energy efficiency in software, hardware and infrastructure. As a result, the same level of demand for digital services and AI is met with a reduced electricity consumption footprint.
- The **Headwinds Case** captures the impact of a downside in the outlook for data centre deployment, particularly due to slower than expected AI adoption. The emergence of local bottlenecks, along with a tight supply chain, causes delays in capacity expansion compared to the most ambitious industry projections.

Given the novelty and technical specificity of data centres as actors in the energy system, the next section presents a brief set of definitions to help readers navigate the rest of the chapter. Readers more familiar with the sector may wish to skip this section and move directly to section 2.2.1, where we present the historical trends in data centre electricity consumption.

2.1.2 Key definitions and concepts

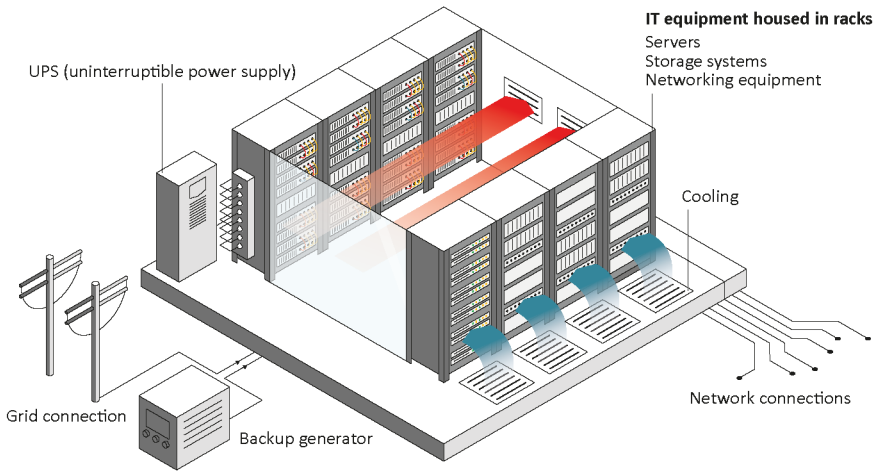
There are several types of data centres. In this report, we use the following categorisations:

- **Enterprise** data centres are run by businesses or institutions for their own use. They are typically smaller and less efficient than other types of data centres and represent around 28% of data centre capacity today. Their share has been steadily decreasing over time, from 85% in 2005.

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- **Colocation and service provider** data centres lease space to customers to house their own computing and storage equipment (colocation) or provide both the space and computing equipment (service providers). Both types of data centres can accommodate hundreds or thousands of customers; an estimated 36% of capacity falls into this category today.
- **Hyperscale** data centres are massive facilities operated by major technology companies, such as Amazon Web Services, Google, Meta and Microsoft. They use scalable, highly efficient infrastructure to support cloud services, web hosting and, increasingly, AI services. Their role has grown quickly from around 10% of data centre capacity in 2010 to 37% today.

Figure 2.1 ▷ Data centre components



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Effective design and integration of key data centre components ensure continuous operation and optimal performance

Data centres are facilities used to house servers, storage systems, networking equipment and associated components that are typically installed in racks and organised into rows (Figure 2.1). This information technology (IT) equipment, and the range of auxiliary equipment required to keep it in working order, comprises the following:

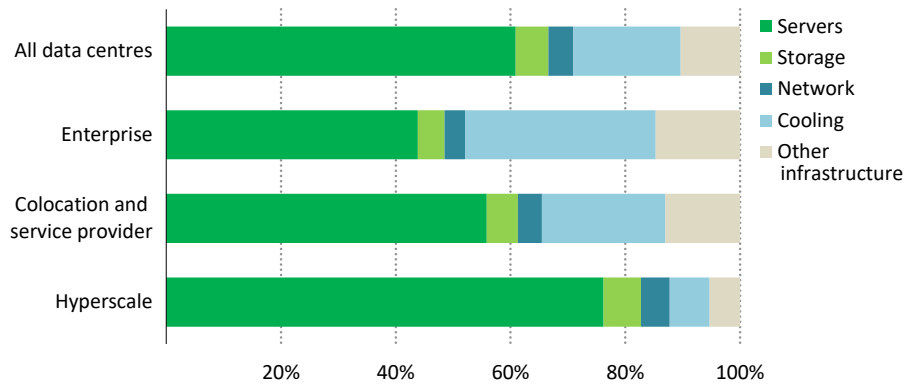
- **Servers** are computers that process and store data. They can be equipped with central processing units (CPUs) and specialised accelerators, such as graphics processing units (GPUs). On average they account for around 60% of electricity demand in modern data centres, although this varies greatly between data centre types (Figure 2.2).
- **Storage systems** are devices used for centralised data storage and backup and account for around 5% of electricity consumption.

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- **Networking equipment** includes switches to connect devices within the data centre, routers to direct traffic and load balancers to optimise performance. Networking equipment accounts for up to 5% of electricity demand.
- **Cooling and environmental controls** consist of equipment that regulates temperature and humidity to keep IT equipment operating at optimal conditions. The share of cooling systems in total data centre consumption varies from about 7% for efficient hyperscale data centres to over 30% for less-efficient enterprise data centres.
- **Uninterruptible power supply batteries and backup power generators** keep the data centre powered during outages. These are rarely used but are necessary to ensure the extremely high levels of reliability that data centres must meet.
- **Other infrastructure** includes lighting and office equipment for onsite staff, etc.

The share of these different components in data centre electricity consumption varies greatly by data centre type, depending on the nature and efficiency of the equipment installed. Figure 2.2 presents the typical breakdown for different data centre types in operation today.

Figure 2.2 ▷ Share of electricity consumption by data centre and equipment type, 2024



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Hyperscale data centres are the most efficient, with the bulk of electricity going to servers and other IT equipment

Several technical characteristics determine data centre electricity consumption:

- **Installed IT capacity** refers to the operating servers, storage and networking devices and is measured in megawatts (MW). **Total installed capacity** includes both IT capacity and the power capacity of auxiliary equipment. In many cases, data centres are only partially filled with servers. **Maximum designed capacity** refers to the maximum capacity of a data centre if it is filled with servers; in many instances, it is smaller than the total installed capacity.

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- The **utilisation rate** of IT equipment measures how much of the available computing resources are actively being used over a given period. Smaller and less-efficient enterprise data centres have average utilisation rates below 20%, while hyperscale data centres with optimised loads can have average utilisation rates of up to 50%.
- **Idle power** is the amount of electricity a device consumes to perform essential background operations when not actively processing workloads. It is typically expressed as a percentage of maximum rated power. A lower idle power is more efficient. Idle power has improved from around 60% in 2010 to around 35% of rated power in modern servers.
- **Power usage effectiveness (PUE)** is the ratio of total facility electricity consumption to the electricity consumption of the IT equipment (PUE = total consumption/IT consumption). It is commonly used as an important indicator of the energy efficiency of a data centre, with a focus on minimising infrastructure electricity consumption (such as cooling and lighting) compared to the electricity consumption of IT equipment. This measure can vary widely from around 2 (meaning 1 kilowatt hour [kWh] of electricity used for cooling and auxiliary equipment for every 1 kWh of electricity used by IT equipment) for enterprise data centres to just under 1.15 for hyperscale data centres (0.15 kWh used for cooling and auxiliary equipment for every 1 kWh used by IT equipment).

Data centre servers deploy several kinds of chips and server architectures:

- **Central processing units (CPUs)** are the primary components of a computer that carry out instructions from programs by performing operations.
- **Graphics processing units (GPUs)** and other "accelerators", such as tensor processing units, are optimised for parallel computations, enabling faster processing of certain tasks.
- **Accelerated servers** are specialised servers equipped with GPUs or similar accelerator chips to enhance computing performance for specific tasks. They are particularly important for AI training and deployment.

2.2 Electricity consumption of data centres

2.2.1 Historical electricity consumption of data centres

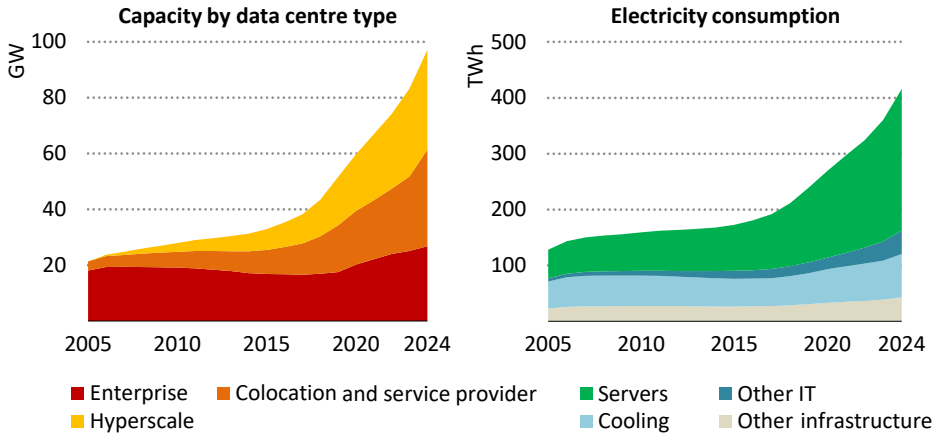
A sharp acceleration in recent years

The Internet revolution took off in the 1990s, and early growth in the demand for digital services was strong. The electricity consumption of data centres in the United States almost doubled between 2000 and 2005, raising concerns about runaway growth (Koomey, 2007). An inflection point occurred around 2007-2008, when slowing growth in data centre electricity consumption indicated a decoupling from the still-booming demand for digital services. Several factors contributed to this slowdown in global data centre electricity

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consumption, including the migration of service demand to more efficient, larger data centres (colocation, service provider and hyperscale), but also continued improvements in hardware efficiency and operating efficiency (declining idle power ratios, for example).

Figure 2.3 ▷ Total data centre electricity consumption by equipment type and data centre type, 2005-2024



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After a decade of limited growth, data centre electricity consumption began to accelerate again after 2015

Note: GW = gigawatt; TWh = terawatt hour.

Sources: IEA analysis based on data from IDC (2024a), OMDIA (2025), and SemiAnalysis (2025).

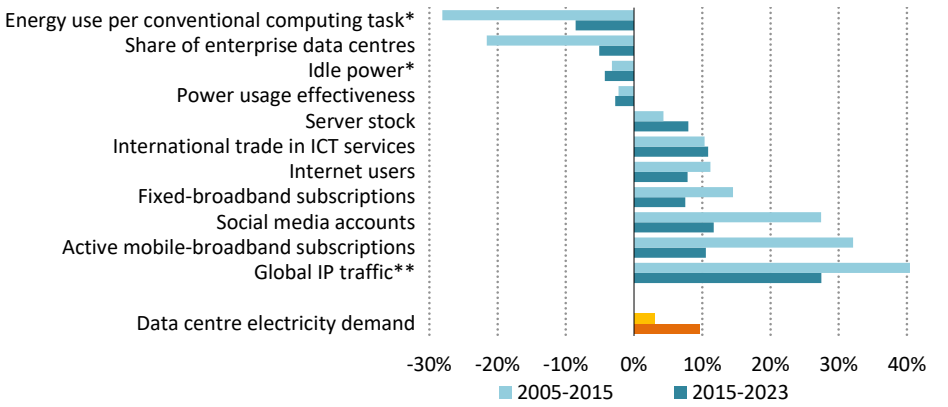
However, a sharp acceleration in data centre electricity consumption took place from around 2017 onwards. Important drivers of this step change were the growth of cloud computing, the shift to online media consumption, the wider use of social media platforms and the rise of AI, which increased the demand for high-performance computing, facilitated by the rise of accelerated servers. Between 2015 and 2024, the capacity of accelerated servers grew four times faster than the total capacity of servers. While accelerated servers are much more efficient on a per-task basis, they also unlocked many new tasks, that were not possible on conventional servers. These new capabilities, among other factors, drove an increase in service demand that outstripped the pace of continued efficiency improvements.

Figure 2.4 provides another view of the drivers of electricity consumption by data centres from 2005 to 2015 and then from 2015 to 2023. From 2005 to 2015, global Internet Protocol (IP) traffic, mobile broadband subscriptions and active social media accounts grew by more than 25% per year. These are proxies for the rapid initial growth in demand for digital services. Growth rates moderated in the period from 2015 to 2023. In contrast, the growth rate of the total stock of servers in data centres accelerated from an annual growth rate of 4% seen in the period 2005 to 2015 to 8% per year from 2015 to 2023. Several key indicators

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of efficiency saw faster improvements from 2005 to 2015, including the rate of the shift from less-efficient enterprise data centres to more-efficient hyperscale, colocation and service provider data centres. As a result of these trends, data centre electricity consumption growth accelerated from 3% annually from 2005 to 2015 to 10% annually from 2015 to 2024.

Figure 2.4 ▷ Average annual change in key drivers of data centre electricity consumption globally, 2005-2015 and 2015-2023



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Robust service demand growth, an acceleration in the total number of servers and a slowdown in some efficiency indicators led to faster electricity consumption growth

* Data starts in 2007. ** Data ends in 2022, estimated for 2022.

Sources: IEA analysis based on data from Cisco (2008), Cisco (2015), Cisco (2019), ITU (2025), Meltwater (2024), SPEC (2024), and World Bank (2024a).

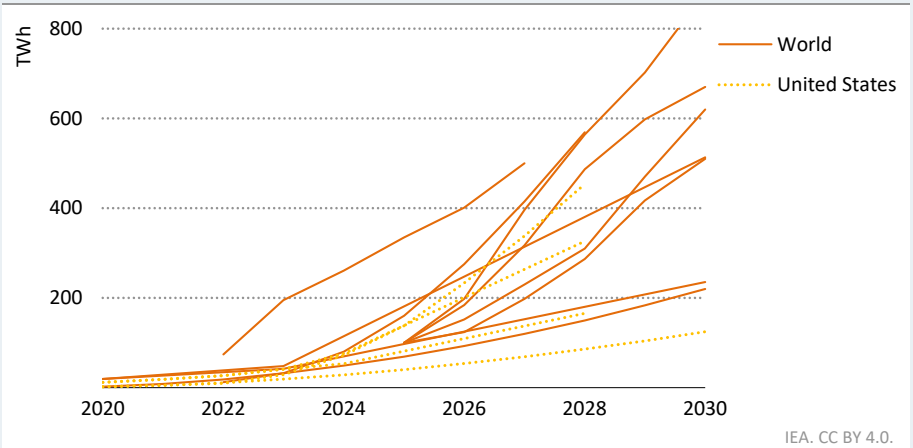
Box 2.1 ▷ What share of data centre electricity demand comes from AI?

How much electricity demand comes from AI specifically is a challenging question to answer. AI is only one of the workloads that run on data centres, and as AI becomes increasingly pervasive, a clear distinction between AI-related and non-AI-related workloads becomes more challenging. There is no comprehensive data on the share of different kinds of workloads, and service providers or colocation data centre operators often have limited visibility over the specific workloads running in their facilities. Moreover, there are often differences in the definition of AI, with some traditional AI sometimes being excluded. In this context, the range of estimates for the share of AI in total data centre electricity consumption is very wide (Figure 2.5).

As a second-best approach, estimates often rely on the electricity consumption of accelerated servers as a proxy for the share of AI in total electricity consumption from data centres. Accelerated servers accounted for 24% of server electricity demand and 15% of total data centre demand in 2024.

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Figure 2.5 ▷ Estimated data centre electricity demand due to AI, 2020-2030



Estimates of the share of AI in total data centre electricity consumption vary widely and are based at best on imperfect proxies

Sources: IEA analysis based on data from Deloitte (2024), Gartner (2024), Goldman Sachs (2024), Schneider Electric (2024), SemiAnalysis (2024), and Shehabi, et al., (2024).

However, this is an imperfect proxy for total AI electricity consumption. AI workloads, especially training, are often run on this type of specialised hardware, but some AI inferencing tasks also run on conventional servers, and some non-AI related tasks, such as high-performance scientific computing, are run on accelerated servers. Looking ahead, some AI inferencing workloads could move from data centres to end-user devices such as mobile phones and laptops (see section 2.4), further increasing uncertainty about AI-related electricity demand.

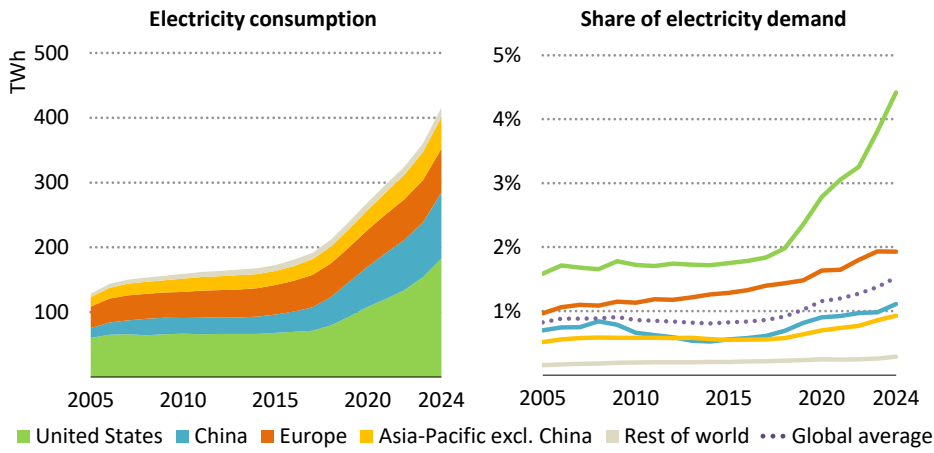
Data centre electricity consumption is not spread evenly around the world

The United States, Europe and China account for around 85% of global electricity consumption from data centres today. In the United States, electricity consumption from data centres grew by around 12% a year between 2015 and 2024. Data centres accounted for around 180 TWh of electricity consumption in 2024 in the United States, nearly 45% of the global total and more than 4% of US electricity consumption from all sources (Figure 2.6).

In China, the data centre sector started to expand significantly from 2015 onwards, with electricity demand growing 15% per year between 2015 and 2024, more than twice the rate seen between 2005 and 2015. Over the same period, electricity consumption across all sectors grew at an annual rate of around 7%. As of today, data centres account for approximately 100 TWh of electricity consumption, roughly equivalent to that of electric vehicles in China. The country accounts for around 25% of global data centre electricity consumption, up from less than 20% a decade ago. However, substantial data gaps make it challenging to accurately estimate China's data centre electricity consumption (IEA, 2025).

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Figure 2.6 ▷ Electricity consumption of data centres by region, 2005-2024



IEA. CC BY 4.0.

The acceleration in data centre electricity consumption observed in 2017 was mainly driven by the United States and, to a lesser extent, by China

Data centres account for slightly less than 2% of Europe's electricity consumption, a share that is higher than China's (1.1%). However, in absolute terms, Europe's consumption is lower, at an estimated 70 TWh in 2024. Europe's share of the global electricity consumption of data centres has decreased over the past decade but still represents slightly above 15%. In Japan, we estimate that data centres account for less than 20 TWh of electricity consumption (about 2% of Japan's total consumption, on a par with Europe). We estimate that data centres account for around 9 TWh of consumption in India (Box 2.2), or about 0.5% of total consumption. However, the sector appears poised for rapid growth.

Box 2.2 ▷ Country focus: India

India has a thriving ICT sector, with the value of IT exports steadily growing to over USD 200 billion in 2024. By comparison, the world's largest oil exporter earned USD 220 billion on export revenues that year. India is also home to around 950 million Internet users. Spurred by data localisation requirements in some sectors, India is now emerging as a rapidly growing data centre market. As of June 2024 India had 2 GW of total installed data centre capacity in operation, together consuming electricity equivalent to 6.5 million Indian households. India's total installed data centre capacity has doubled in only four years, and over 2 GW of further maximum designed capacity is in the pipeline and planned to come online over the next two years. This means that total installed capacity is on track to reach nearly 5 GW by 2030 (Figure 2.7).

The government's IndiaAI Mission, with a budget of USD 1.2 billion, consists of several objectives, including the development of an AI computing ecosystem with over 18 000 GPUs to support AI start-ups and research. In addition, there are incentives from

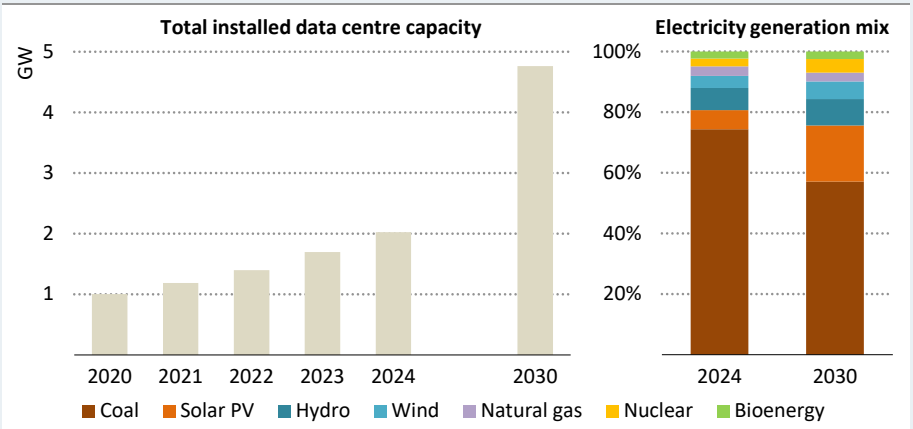
IEA. CC BY 4.0.

state governments for data centres; for instance, Uttar Pradesh announced a 100% exemption on electricity duty and transmission charges for ten years for new data centres.

Electricity consumption from data centres is contributing to India's electricity demand growth at a time when India is already among the world's fastest-growing electricity markets. Coal fuels 74% of electricity generation in India today, providing much of the firm power to the grid, and the dominance of coal in the mix is likely to continue beyond 2030. However, since India's "open access" rules enable the direct purchase of power from generators, several technology companies are signing power purchase agreements (PPAs) directly with renewable energy generation companies to reduce their emissions. For example, the data centre subsidiary of Indian telecommunications major Bharti Airtel announced it would procure 140 GWh of renewable energy annually and has been working with generation companies to set up captive solar photovoltaic (PV) and wind capacity for their data centres.

To ensure that the upcoming wave of new data centre construction remains on target, India will need to address long-standing issues of grid reliability to capitalise on data centre and AI growth. In the current context, backup and captive power generation for data centres remains a critical consideration owing to the risk of power supply interruptions from the grid. Grid infrastructure creation and upgrades will also need to keep track of new data centre construction. Data centres are proving to be important energy consumers in India, creating additional demand for power generation, notably from solar PV and wind, and driving investment in power backup options (including battery storage) and transmission infrastructure upgrades.

Figure 2.7 ▷ India's total data centre capacity and electricity generation mix, 2020-2030



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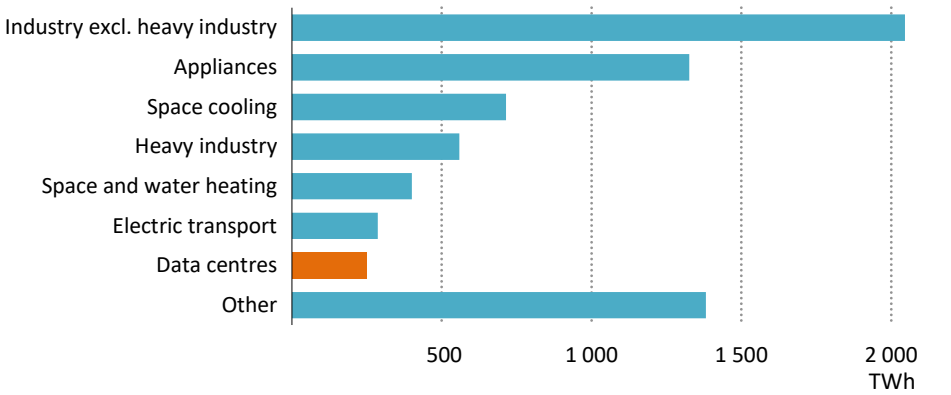
Data centre total installed capacity in India is set to double by 2030; while coal dominates the electricity mix in India, the share of renewables increases to 35% by then

IEA. CC BY 4.0.

Data centre electricity consumption from a broader perspective

With the rapid growth seen in recent years, the data centre sector accounted for around 4% of global growth in electricity consumption between 2014 and 2024. However, other drivers, such as growing appliance adoption in buildings and industrial electrification, were more significant. Data centres accounted for around 250 TWh of incremental electricity consumption in this period, roughly equivalent to the electricity demand of Spain. In comparison, the electricity consumption of space cooling grew by around 700 TWh (Figure 2.8).

Figure 2.8 ▻ Increase in electricity demand by sector in the Base Case, 2014-2024



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Over the past decade, growth in electricity demand for data centres increased by almost as much as for transport

All historical data are estimates, not measurements

Currently, very few governments mandate reporting and publication of comprehensive statistics on data centre electricity consumption (see Chapter 5). As a result, all data on the historical consumption of data centres at the global level are the result of estimates based on a variety of sources. These sources come with different challenges and gaps; combined with the lack of common definitions (Masanet, Lei and Koomey, 2024), this results in widely divergent estimates, even for historical consumption.

Often, companies and institutions that track or report on data centre capacity, including real estate companies, consulting companies and data centre operators, do not use a consistent scope or definitions. A key distinction is between the maximum designed capacity and the actual installed IT capacity. Data centres are often not filled to their maximum designed capacity, and IT capacity is usually ramped up progressively in new data centres. Access to detailed data on the installed capacity of data centres requires subscriptions to expensive third-party data services, an option that is often not available to many actors.

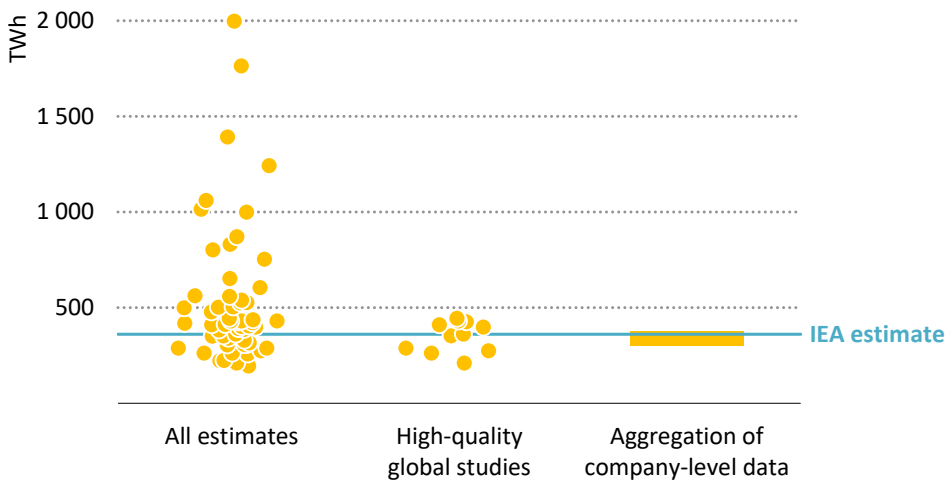
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Several recent studies rely on outdated or overly simplistic PUE assumptions due to a lack of high-quality, publicly available data for different data centre types and regions. The load factor of data centres is also marked by uncertainty. This metric is determined by IT equipment utilisation rates and idle server power consumption, both of which vary greatly across different workloads and hardware configurations. Combined with a lack of available data, this complicates aggregate load factor estimations, leading to significant discrepancies in consumption estimates.

The rapid adoption of GPUs and other accelerated server designs further compounds these challenges. There are limited available data on annual shipments and the installed stock of accelerated servers. This has a large impact on demand estimations, as accelerated servers are much more power intensive than conventional servers. Finally, companies operating in the sector, including hyperscale and colocation providers, largely do not report their data centre electricity consumption specifically.

As a consequence, all historical data regarding global data centre electricity consumption are modelled estimates, not measured data, and the range of estimates is wide (Figure 2.9). Triangulating multiple data sources does lead to broadly converging estimates that align with the IEA estimates provided in this chapter (Kamiya and Coroamă, 2025a). However, the process is intensive in time, resources and expertise. The data and methodological annex to this report provides more details on the methodology used by the IEA to estimate data centre electricity consumption from data centres.

Figure 2.9 ▷ Comparison of three approaches to estimating global data centre electricity consumption, 2023



Different modelling approaches can lead to a wide range of estimates

Source: IEA analysis based on data from Kamiya and Coroamă (2025a).

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2.5 Electricity supply to meet data centre demand

2.5.1 Procurement strategies of technology companies

Procuring electricity supplies that are reliable and cost-effective is crucial to meeting the rapidly growing electricity demand from data centres. Many technology companies and large data centre operators have set ambitious goals for reducing emissions and procuring clean energy (Table 2.2). To meet these objectives, data centre operators use various procurement strategies. These vary by company and region, with liberalised electricity markets generally offering more procurement choices than regulated markets. In addition to sourcing the grid electricity mix, procurement strategies include acquiring electricity through PPAs. Many companies also purchase renewable energy certificates to meet their clean energy targets.

The recent surge in data centre electricity demand has led to significant interest in additional natural gas-fired power generation, largely in the United States, where natural gas is a low-cost fuel. Gas turbine manufacturers are reporting an uptick in orders, and several large data centre operators have announced partnerships with utilities and energy companies developing new gas-fired power capacity. In Louisiana, for example, Entergy Louisiana is planning more than 2 GW of additional gas-fired power generation to provide power for Meta data centres. NextEra Energy and GE Vernova also aim to develop natural gas-fired power generation projects across the United States, primarily to meet the growing electricity demand of data centres. At the same time, many US utilities are currently revising their integrated resource plans to account for rising data centre electricity demand, proposing additional natural gas-fired capacity to meet it. To bring down emissions, some data centre operators are considering fitting natural gas-fired plants with carbon capture in the long run.

Most renewable energy PPAs are financial agreements for annual volumes of electricity and are not tied to the hour-to-hour consumption profile of a data centre or the generation profiles of the renewable assets, which can also be located in different regions. While these PPAs help data centre operators meet their clean energy targets, the separation of renewable generation and data centre consumption often means other sources, like natural gas or coal, are used to meet physical electricity needs. This results in a physical electricity mix that differs from the procured, or “financial”, electricity mix.

To enhance their sustainability strategies and further support decarbonised grids where they operate, some technology companies are concluding PPAs with hourly matching. This means that some or all of their electricity consumption is matched hour-by-hour by a portfolio of renewable energy and storage assets, or other types of low-emissions power generators located in the same region. For instance, Google seeks to achieve hourly matching, and Microsoft has signed hourly PPAs in support of its goal to be carbon negative by 2030. In order to achieve these goals, they are moving to deploy portfolios of renewable energy and storage projects that can increase hourly matching, as well as signing PPAs with dispatchable sources of low-emissions electricity, such as hydro, nuclear, geothermal or natural gas with carbon capture. In September 2024, for example, Microsoft and Constellation Energy concluded a 20-year PPA for the restart of the Three Mile Island nuclear plant.

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Table 2.2 ▷ Emissions reduction and clean energy targets of corporate data centre operators

Company	Estimated data centre capacity (MW)	Net zero emissions target year	Corporate clean, green or renewable electricity target*	Current share	Hourly matching target*
Meta	9 780	2030	100% renewable since 2020	100%	
Google	8 960	2030	100% renewable since 2017	100%	100% by 2030
Amazon	7 660	2040	100% renewable since 2023	100%	
Microsoft	6 970	2030	100% renewable by 2025	100%	100% by 2030
Digital Realty	2 740			66%	
Equinix	1 850	2030	100% renewable by 2030	96%	
Tencent	1 760	2030	100% green by 2030	12%	
Alibaba Cloud	1 660	2030	100% clean by 2030**	56%***	
Aligned	1 290	2040	100% renewable since 2020	100%	
Huawei	1 260	2040		> 50%	
Apple	1 240	2020	100% renewable since 2018	100%	
Vantage	1 180	2030		58%	
CyrusOne	1 120	2030	100% carbon-free energy by 2030	62%	
NTT Data	1 110	2035	100% renewable by 2030**	49%	
QTS Data Centers	1 060			65%****	
Baidu	980	2030		5%	
GDS	980	2030	100% renewable by 2030	36%	
Chindata	900	2060	100% renewable by 2040**	7%	
Switch	660	2021	100% renewable since 2016	100%	
Princeton Digital	620	2030	100% green by 2030	14%	

* Only targets with specified years are included. ** Target covers data centres only. *** Percentage of clean electricity consumed at Alibaba Cloud's self-built data centres. **** Percentage of low-emissions electricity utilised by QTS' facilities.

Notes: Data centre operators are ranked by their total estimated data centre capacity as of the end of the first half of 2024, based on OMDIA (2025). The OMDIA database may not be complete but was used to provide a consistent source across diverse companies for installed capacity. The net zero targets are for Scope 1 and Scope 2 emissions.

As part of these strategies, technology companies are also supporting the development and commercialisation of innovative low-emissions baseload technologies, such as small modular reactors (SMRs) and next-generation geothermal. To date, plans to build up to 25 GW of SMR capacity associated with supplying the data centre sector have been announced worldwide, almost all of them in the United States, although projects are at varying stages of maturity and certainty. The first projects are expected to start to materialise only towards the end of this decade.

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Advances in geothermal technology, including horizontal drilling and hydraulic fracturing pioneered by the oil and gas industry, are promising to increase the number of locations in which geothermal energy could be harnessed to provide a cost-competitive source of baseload electricity. Google has partnered with Fervo Energy, which developed a 3.5 MW first-of-its-kind next-generation geothermal power pilot project in Nevada. This project started feeding electricity into the grid in November 2023. In June 2024, Google and NV Energy, Nevada's utility, entered into a power supply agreement for Fervo's 115 MW Corsac next-generation geothermal project, which is currently under development. Meta has signed an agreement with Sage Geosystems for 150 MW of capacity to power its data centres from 2027, while Microsoft and partner G42 are planning the construction of a data centre campus powered by geothermal power in Kenya.

As an alternative to procuring electricity from utilities or through PPAs, some technology companies are co-locating data centres with power generation facilities, enabling them to generate some or most of their own electricity directly. The primary benefit of co-locating generation is potentially faster development times, as this approach can allow them to downsize or opt for an interruptible grid connection, saving costs and helping to alleviate grid congestion. The downsides are higher complexity, increased permitting requirements, higher investment costs, potentially lower reliability and a greater maintenance burden.

Recent years have seen rising interest in co-locating data centres and generation assets. Google is partnering with Intersect Power and TPG Rise Climate to develop co-located clean energy projects with data centres, aiming for completion by 2027. Chevron and Engine No. 1 are partnering with GE Vernova, with plans to supply up to 4 GW of natural gas capacity to co-located data centres, aiming to start operations by the end of 2027. Amazon and Talen Energy have signed a 10-year PPA for 300 MW to 960 MW of nuclear energy from the Susquehanna nuclear plant to supply a co-located data centre, although a recent Federal Energy Regulatory Commission ruling on the repurposing of existing grid-connected power plants to directly provide power to co-located loads halted plans to expand the electricity supply beyond the initially awarded 300 MW. The commission has recently initiated a process to examine the colocation policy.

Box 2.4 ▻ Data centre operators are leading the corporate PPA push

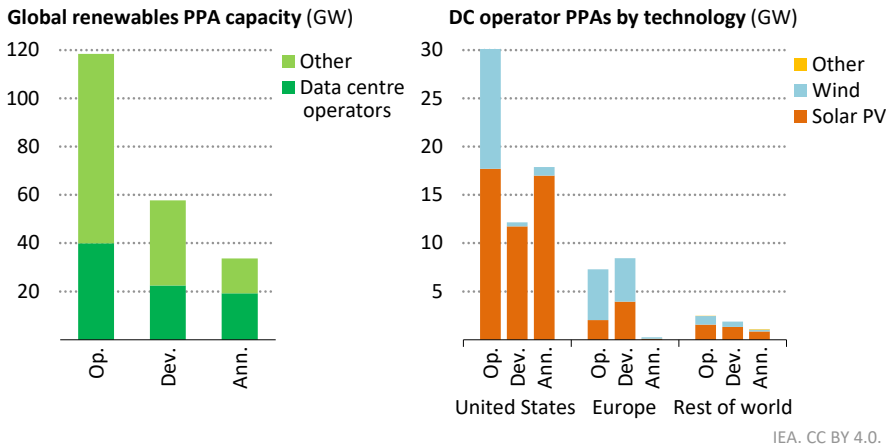
A PPA is a long-term contract in which an electricity generator sells power to a buyer at a fixed price for a specified period. In regulated markets, green tariffs can serve a similar role, with a utility acting as an intermediary. To date, nearly 120 GW of operational renewables capacity has been procured through corporate PPAs globally (Figure 2.16). Technology companies operating data centres account for over 30% of this capacity. In 2024, technology companies' renewables PPAs were sufficient to cover roughly 20% of the estimated 415 TWh of global electricity demand from data centres.

An additional 60 GW of PPA-related capacity is currently under development – meaning projects for which financing and/or permits have been secured or which are under

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construction. Almost 40% of this capacity is contracted by data centre operators. Data centre operators have also been responsible for the lion's share of recent announcements, accounting for almost 60% of the 34GW of the renewables capacity for which corporate PPAs have been announced but which has not yet entered the development stage. Projects that are under development or have been announced would provide sufficient capacity to cover approximately 15% of the projected electricity demand growth from data centres to 2030.

Figure 2.16 ▷ Global renewables capacity contracted through corporate PPAs by development status, offtaker and technology



Data centre operators account for over 30% of active PPAs and the majority of announced PPAs

Notes: Op. = operational; Dev. = under development; Ann. = announced; DC = data centre; PPA = power purchase agreement. The cut-off date is February 2025. Only individual known projects are considered. Other includes bioenergy and geothermal.

Source: IEA analysis based on data from BNEF (2025).

Of the operational renewables PPA capacity contracted by data centre operators, 75% is located in the United States, with nearly 20 GW of solar PV and about 12 GW of onshore wind under contract there, followed by Europe with 20%. Over 50% of the capacity under development is also located in the United States – almost of all of it solar PV, while Europe accounts for around 35%. In the European Union and United Kingdom, offshore wind farms account for most of the under-development capacity contracted by data centre operators in the region. Announcements for additional PPAs have so far focused mostly on the United States, with nearly 90% of the announced capacity. While there has recently been an increase in announcements from other parts of the world, including Southeast Asia and India, significant regulatory hurdles continue to limit the deployment of PPAs, in particular in emerging market and developing economies.

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2.5.2 Matching electricity supply with data centre demand

Electricity supply to meet data centre demand can come from a wide set of sources, each with unique characteristics related to technical performance, cost, emissions, the development process and lead times. Consideration of these options, either to be developed onsite or connected through the grid, is critical to scaling up electricity supply to meet data centre demand.

Table 2.3 ▷ Sources of electricity to match the needs of data centres

Electricity source	Construction period	Variable or dispatchable	Global average CO_2 intensity ($g\ CO_2/kWh$)	Global average LCOE (USD/MWh)
Utility solar PV	1-4 years	Variable	0	60
Wind onshore	2-5 years	Variable	0	50
Wind offshore	3-7 years	Variable	0	110
Hydropower plant	5-15 years	Variable (run-of-river) Dispatchable (reservoir)	0	80
Conventional geothermal	3-8 years	Dispatchable	0	80
Nuclear (new)	5-15 years	Dispatchable	0	90
Nuclear (restart)	2-5 years	Dispatchable	0	60
Coal	3-6 years	Dispatchable	960	80
Gas CCGT	2-4 years	Dispatchable	390	80
Gas GT	1-3 years	Dispatchable	620	220
Grid connection	3-7+ years	Dispatchable	United States: 350 China: 600 Southeast Asia: 610 Europe: 240 World: 460	-

Notes: CO_2 = carbon dioxide; $g\ CO_2/kWh$ = grammes of carbon dioxide per kilowatt hour; CCGT = combined-cycle gas turbine; GT = gas turbine; LCOE = levelised cost of electricity; MWh = megawatt hour. Construction period refers to typical projects, excluding supply chain equipment delays. Average emissions intensity is assessed on direct emissions from the average mix between 2021 and 2023. Other assumptions come from the *WEO-2024* (IEA, 2024). Nuclear (new) includes small modular reactors.

As data centres are projected to grow rapidly over the years to come, the strategy to build out and ensure a stable and efficient source of electricity becomes crucial. Currently, the only reliable electricity sources that can be developed within a short timeframe – ideally one to two years (Table 2.3) – are solar PV and gas turbines, aligning with the typical construction timeline of data centres. Even in these cases, supply chain delays or tight supplies can further extend development times (Box 2.5). Wind turbines could also be a viable option in terms of deployment speed; however, lengthy permitting processes often extend their timeline to around five years, a similar development time to conventional geothermal, or longer. Other dispatchable technologies, such as large-scale nuclear reactors or hydropower plants,

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typically require closer to a decade or more to complete. Once SMRs or next-generation geothermal become commercial, they may also offer medium-length development times of approximately three to five years.

Box 2.5 ▻ Will strained supply chains slow down the growth of natural gas-fired power generation?

Low fuel costs, high reliability and the ability to operate at high load factors make natural gas-fired generation an attractive option for data centre operations in the United States, in particular. Several gigawatts of new capacity targeting the data centre market have been announced by developers in late 2024 and early 2025. These announcements come on top of the substantial volumes of new gas-fired capacity planned by utilities to meet overall electricity demand growth.

As a result, orders for new gas turbines from utilities and project developers have surged over the past two years. However, this sudden increase in orders is hitting a global gas turbine supply chain that has seen limited investment in manufacturing capacity due to years of stagnant electricity demand in advanced economies and a recent slowdown in global additions of gas-fired power. Global additions of natural gas plants peaked at nearly 110 GW in 2002, have averaged around 60 GW per year since then and have fallen to an average of 40 GW per year since 2020.

Three main manufacturers – GE Vernova, Siemens Energy and Mitsubishi Power – supply turbines for about two-thirds of the gas-fired power plants currently under construction globally and are reporting growing backlogs. Turbine deliveries for new power plants now face delays of several years in many instances. The uptick in activity has meant that other elements of supply chains, including labour and other goods, are also tight, potentially delaying the commissioning of new gas-fired power plants beyond 2030.

These extended delivery timelines cast doubt on the ability of utilities and energy companies to scale up natural gas-fired generation as quickly as planned to meet rising demand, especially in the near term. They are also driving up capital costs for the developers of new gas-fired plants. High demand and constrained supply increase the pricing power of the turbine manufacturers. Longer delivery timelines lead to increased financing costs and can disrupt construction schedules, increasing the risk of cost overruns. Consequently, some developers are opting to pay premiums to move to the front of the queue for turbine deliveries.

The strained supply chain is also affecting existing plants. Servicing activity and new unit production compete for factory capacity, and as manufacturers prioritise the production of new turbines, this reduces the availability of refurbishment capacity and component parts, raising the risk of plant outages. Additionally, the increased demand has driven up the cost of new long-term gas turbine maintenance contracts, raising plant operating expenses.

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The projected growth in electricity demand means that there is a need for additional secure capacity to ensure the reliability of the electricity supply. In the United States, the revised integrated resource plans of the country's utilities call for an additional 84 GW of natural gas-fired capacity by 2035 (see Box 2.6 for more details). The limited availability of gas turbines may require utilities and project developers to explore alternatives to new natural gas-fired plants to address near-term growth in the demand for electricity and secure capacity. This includes upgrading existing plants to enhance their electrical output, although tight supply chains have seen lead times for such measures go up as well. Technical improvements, such as retrofitting better turbine blades, water injection and inlet air cooling, can increase the efficiency and raise the capacity of simple open-cycle gas turbines by 3-10%. If applied to the United States' fleet of existing open-cycle gas turbines, such refurbishments could provide about 4 GW to 15 GW of additional capacity. If sufficient space is available at the site, open-cycle gas plants can also be upgraded to combined-cycle through the addition of a heat recovery steam generator and steam turbine. A combined-cycle gas turbine power plant can produce up to 50% more electricity from the same amount of fuel.

Technology costs are another important factor in considering supply options to meet data centre demand. Wind and solar PV technologies are currently among the cheapest sources of electricity. Additionally, in regions where natural gas prices are low, such as the United States and the Middle East, gas turbines offer an alternative. To be comparable with dispatchable sources of electricity, solar PV and wind need to be paired with storage to increase their availability throughout the day, but the cost comparison remains valid. Coal-fired power can be one of the lowest-cost sources of electricity in places where prices on CO_2 emissions are low or zero, but development times for coal plants can be quite long outside China.

Emissions at the point of electricity generation are an important factor, especially in light of the sustainability targets set by many technology companies and national and international climate goals. Coal-fired power has the highest emissions intensity of the potential options (oil-fired power is of a similar level), with natural gas-fired power plants emitting roughly half as much CO_2 per unit of electricity output. Excluding indirect emissions from their life cycle – such as extraction, manufacturing and decommissioning – renewable energy and other low-emissions sources like nuclear energy have no direct CO_2 emissions.

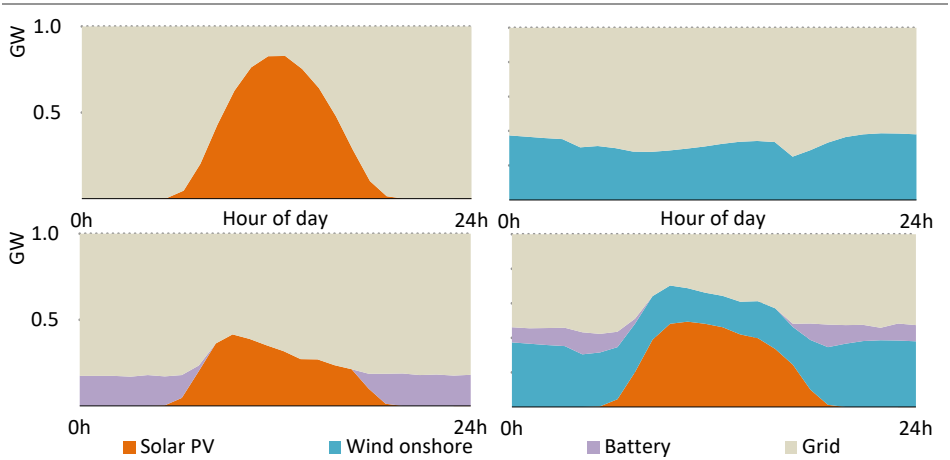
Hourly matching: What does it really take?

The most common arrangements for procuring renewable electricity are based on annual volume matching. Annual matching means that enough capacity is procured to meet 100% of the user's electricity demand over the course of the year, without consideration of precisely when demand and supply occur. Conventional "annual matching" PPAs can help drive the installation of new renewables capacity. However, hourly matching of low-emissions electricity PPAs ensures that electricity consumption in each hour of the year is met by low-emissions energy sources.

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Dispatchable sources of electricity generation, such as hydro, geothermal and nuclear, can generally match hourly demand throughout the year, but this is not the case with variable renewables. For example, where solar PV alone is procured to cover 100% of annual demand, the share of hourly demand covered can average 35-45%. Effectively this means that on an hourly basis, solar PV output is only able to meet 35-45% of data centre demand due to its output profile. In the hours when the procured solar PV is above data centre demand, the excess can be available to the grid and other consumers. In the hours when solar PV output is below hourly data centre demand, the remaining demand must be met by other sources. Again, this results in a physical electricity mix that differs from the procured, or "financial", electricity mix. The associated CO_2 emissions for the electricity supply to meet data centre demand depend on the extent to which low-emissions sources cover data centre demand and the emissions intensity of the grid electricity.

Figure 2.17 ▷ Daily average electricity generation profiles of wind, solar PV and battery storage to meet baseload demand in Virginia, United States



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Renewables, coupled with storage, can meet a flat demand profile

Note: The graphs depict four different use cases considering 1.5 GW of solar PV, wind or both, and a 1 GW battery with 4 GWh of storage.

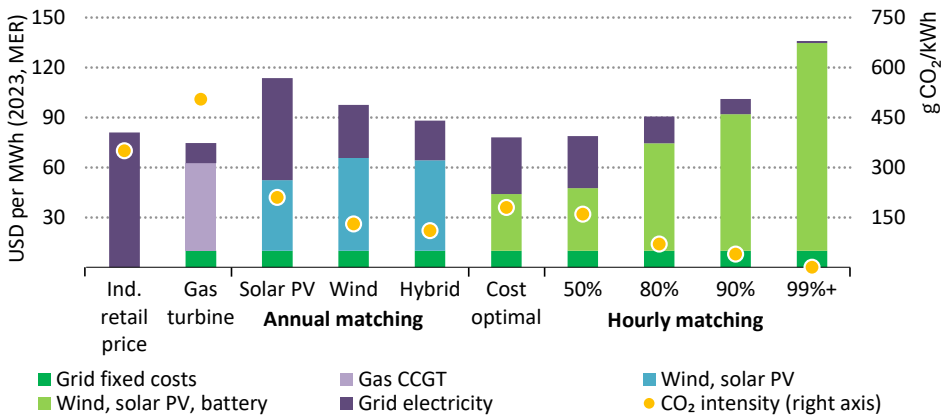
Hourly matching of the procured electricity supply to the data centre electricity demand is an approach pursued by several large technology companies, but achieving this ambition with variable renewables comes with challenges. Solar PV and wind generation are inherently variable. Solar PV varies across the day and seasons. Wind production is less variable on average but can vary quickly from hour to hour, with extended periods of low or high generation. However, hybrid projects combining solar PV, wind and storage offer a better match to baseload demand, with storage helping to smooth out variable output from

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renewables. Solar PV combined with battery storage has the advantage that it can be deployed quickly and provides a more constant supply. Combining solar PV, wind and battery storage results in an even flatter supply (see Figure 2.18). To align with baseload demand on an hourly basis, the installed capacity of renewable sources must be higher than the average demand.

In order to analyse the ability of solar PV, wind and battery storage to meet baseload demand, a new analysis of over 1 000 use cases covering eight configurations in more than 100 regions was carried out. The regions include European countries, each state of the United States and each province of China. Different procurement strategy configurations were tested, resulting in different combinations of renewable and storage technologies. The remaining portion of electricity demand not covered by renewable sources was assumed to come from the grid at the average industry retail price. The analysis measures the hourly matching of supply and demand, the average cost and the associated CO_2 emissions.

Figure 2.18 ▷ Average cost of electricity consumed by component for different portfolios and average CO_2 emissions intensity in the United States, 2025



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80% hourly matching low-emissions portfolios are comparable in costs with annual volume matching projects in the United States.

Notes: MER = market exchange rate; Ind. = industrial; CCGT = combined-cycle gas turbine. Annual matching = portfolio of wind and/or solar PV optimised to meet 100% of annual demand; 50% hourly matching = portfolio of wind, solar PV and batteries to reach at least 50% hourly matching with demand; 80% hourly matching = to reach at least 80% hourly matching with demand; 90% hourly matching = to reach at least 90% hourly matching with demand; 99%+ hourly matching = to reach at least 99% hourly matching with demand. Assumptions for the industry retail price of electricity are taken from 2023 historical data. Assumptions for gas turbine costs are based on a natural gas price of USD 19/MWh and an 85% capacity factor for CCGT technology with 60% efficiency. Assumptions for capital expenditures and operational expenditures for solar PV, onshore and offshore wind, and battery storage are taken from IEA (2024) for the year 2025.

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In the “annual match” configuration where solar PV alone is procured to cover 100% of annual demand, the share of hourly demand covered averages about 40% in the United States in most locations (see Figure 2.18). Where onshore wind alone is procured, hourly coverage averages almost 65% but ranges from 55% to 75% depending on the US state. In Europe and China, solar PV alone covers an average of 40%, with a wide range across countries and provinces, and wind alone covers similar shares to the United States. In China the range across provinces is wider for wind alone, ranging from 40% to 80%.

We assessed the optimal sizing of solar PV, onshore wind and hybrid projects to minimise costs while meeting annual electricity demand in volume (referred to as the “hybrid annual matching” configuration). For hybrid projects that combine wind and solar PV to meet annual electricity demand, the average hourly coverage share is 70%, with a range from 40% to 80% across different states of the United States. In Europe, the share of demand covered by renewables in the hybrid annual matching configuration averages 70% and ranges from 40% to 85% in countries with the best renewable potential. In China, the annual matching configuration for hybrid projects usually covers 65% of baseload demand, ranging from 40% to almost 85% across provinces.

We also explored the cost optimal configuration of wind, solar PV, battery storage and purchasing of grid electricity. The assessment is based on resource potential and grid electricity costs, without specific constraints on volume or demand coverage. In this “cost optimal” configuration, renewables cover an average of 50% of demand in the United States, ranging from 25% to 70% in the most resource-rich states.

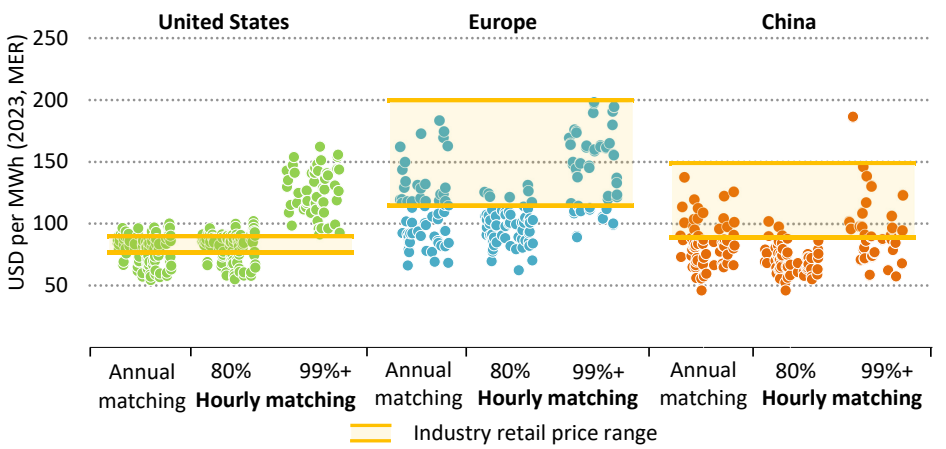
The last portfolios focused on the optimal sizing of wind, solar PV and battery storage portfolios to achieve a specific target of hourly matching between renewables supply and baseload demand. The analysis finds that ensuring 80% hourly matching of renewable sources with baseload demand is comparable in cost to the annual matching configuration in the United States, with the added benefit of guaranteeing 80% hourly matching with low-emissions sources. This 80% guarantee configuration aligns with the grid retail price in the United States at USD 80 per megawatt hour (MWh) (without including grid fixed costs like connection charges). Achieving nearly full hourly matching with hybrid projects adds a premium to overall costs, over 50% above the grid electricity price for industry in the case of the United States, because of the required additional capacity in both supply and storage. However, a higher share of hourly matching reduces exposure to electricity market price volatility, protecting consumers from high prices.

Looking across regions, we find several similar results, including that 80% hourly matching portfolios are comparable in cost and even more affordable than annual matching hybrid projects. Annual matching hybrid projects can be more expensive because of their lack of storage and greater reliance on grid electricity. In many countries in Europe and provinces in China, the respective average costs of USD 100/MWh and USD 70/MWh for the “80% hourly matching” configuration are below the 2023 average industry retail electricity price.

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The analysis also revealed several regional differences. In the cost optimal configuration for Europe, the share of demand covered by wind, solar PV and battery averages 80%, which is notably higher than for the United States due to a generally more expensive electricity price. In China, the coverage share averages 70% and ranges between 30% and more than 90% in some provinces in the most cost-effective case because of the lower investment costs compared to the United States. When full hourly matching is the target, the cost premium in China is 5% above the 2023 average grid cost. In Europe, 90% hourly matching can be achieved for USD 105/MWh and 99% coverage for less than USD 150/MWh on average (see Figure 2.19).

Figure 2.19 ▷ Total cost of electricity per unit consumed for hybrid options of wind, solar PV and battery in the United States, Europe and China



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Hybrid wind, solar PV and battery portfolios can meet 80% of baseload demand at an average cost competitive with industry retail prices in the United States, Europe and China

Notes: MER = market exchange rate. Annual matching = portfolio of wind and solar PV optimised to meet annual volume demand target; 80% hourly matching = portfolio of wind, solar PV and batteries to reach at least 80% hourly matching with demand; 99%+ hourly matching = to reach at least 99% hourly matching with demand. Each dot represents a different use case based on the renewable potential of various locations in the United States, Europe and provinces in China. Assumptions for the industry retail price for electricity are taken from the minimum and maximum values across historical data between 2014 and 2023 for each country in Europe, US state and province of China. Assumptions for capital expenditures and operational expenditures for solar PV, onshore and offshore wind, and battery storage are taken from IEA (2024) for the year 2025.

Constant baseload demand for data centres does not necessarily imply conventional dispatchable power sources. As variable renewables are now cheaper and faster to deploy in many regions compared to other technologies, pairing them with storage can increase their alignment with baseload-type demand. Hybrid portfolios of wind, solar PV and storage can cover a relatively high share of demand on an hourly basis at a competitive price. Aiming for a very high share of hourly matching raises the costs, which can exceed the average

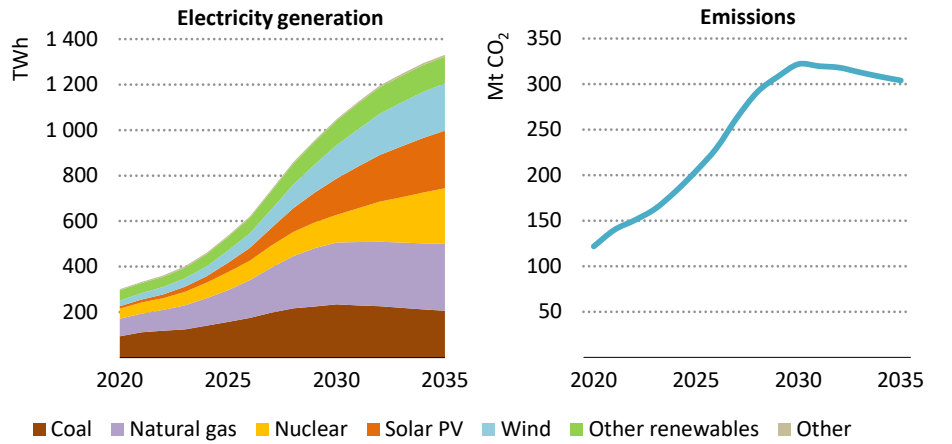
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industry retail price depending on the region. Compared with conventional annual matching PPAs, hourly matching PPAs with a high share of low-emissions sources provide a higher guarantee of covering electricity demand, reducing CO_2 emissions and mitigating the volatility risk associated with electricity prices. The role of renewables should also be analysed at the broader system level to better assess the balance of the variability.

2.5.3 Electricity supply in the Base Case

Global electricity generation to supply data centres is projected to grow from 460 TWh in 2024 to over 1 000 TWh in 2030 and 1 300 TWh in 2035 in the Base Case. Renewables meet nearly half of the additional demand to 2030, followed by natural gas and coal, with nuclear starting to play an increasingly important beyond 2030 (Figure 2.20).

Figure 2.20 ▷ Global electricity generation for data centres and the associated CO_2 emissions in the Base Case, 2020-2035



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Between now and 2030, renewables meet nearly half of the increase in global data centre electricity demand, followed closely by natural gas and coal-fired electricity generation

Coal, with a share of about 30%, is the largest source of electricity, though this varies significantly by region, with the highest contribution found in China. Renewables – primarily wind, solar PV and hydro – currently supply about 27% of the electricity consumed by data centres globally. Natural gas is the third-largest source today, meeting 26% of the demand, followed by nuclear with 15%. It should be noted that this analysis considers the fuel mix of the electricity physically consumed by data centres (considering both onsite generation and electricity received through the grid, taking into account the fuel mix of the local electricity systems they are located in) rather than the contractual mix of different data centre operators.

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Taken together, renewables remain the fastest-growing source of electricity for data centres, with total generation increasing at an annual average rate of 22% between 2024 and 2030, meeting nearly 50% of the growth in data centre electricity demand. This growth is primarily driven by the rising deployment of wind and solar PV in power systems across the globe, with some of the new capacity financed through PPAs with technology companies. Some data centre operators also invest directly in co-located renewables. Even so, new demand from data centres is a significant near-term driver of growth for natural gas-fired and coal-fired generation, through both higher utilisation of existing assets and new power plants. Natural gas and coal together are expected to meet over 40% of the additional electricity demand from data centres until 2030. After 2030, SMRs enter the mix, providing a source of baseload low-emissions electricity to data centre operators. Currently, hyperscalers are among the key corporate backers of SMR development. Coupled with the ongoing growth of renewable electricity generation, the resulting increase in nuclear electricity generation leads to an absolute decline in coal-fired generation for data centre operations by 2035. Consequently, CO_2 emissions from electricity generation for data centres peak at around 320 Mt CO_2 by 2030, before entering a shallow decline to around 300 Mt CO_2 by 2035. Despite rapid growth, data centres remain a relatively small part of the overall power system, rising from about 1% of global electricity generation today to 3% in 2030, accounting for less than 1% of total global CO_2 emissions (see Chapter 5 for more details).

Regional outlook

The United States and China are by far the largest data centre markets today. In both countries, most of the electricity consumed by data centres is produced from fossil fuels, which also meet most of the increase to 2030. However, the rising deployment of renewables, and later nuclear, is expected to slow the growth of fossil fuel power generation after 2030.

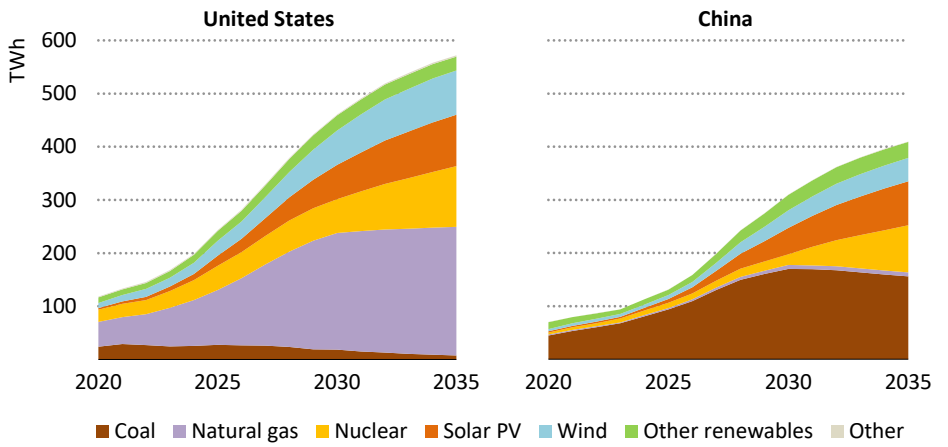
With a share of over 40%, natural gas is currently the biggest source of electricity for data centres in the **United States**, followed by renewables – mostly solar PV and wind – at 24%, as well as nuclear and coal power with shares of around 20% and 15%, respectively. As demand growth is particularly rapid over the next five years, natural gas is the largest source of additional supply, adding over 130 TWh of annual generation until 2030. Utilities are revising their integrated resource plans, with the construction of additional gas-fired power plants planned across the country, some of them to support the increase in data centre loads (see Box 2.6). Furthermore, some data centre operators are partnering with utilities and energy companies to expand gas-fired capacity, some of it directly co-located with data centres. Renewables are the second-largest source of additional electricity supply, adding 110 TWh to data centre electricity supply between 2024 and 2030. This is mainly due to the continuing increase in the share of wind and solar PV in the electricity mix of most states, as well as some data centre operators investing in co-located renewables.

Nuclear power plays a significant role in meeting data centre electricity demand in the United States, particularly after 2030 when the first SMRs are expected to be commissioned.

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Technology companies have plans to finance more than 20 GW of SMRs to date, though successful development of the technology could open up even larger opportunities. Together with the ongoing increase in renewable electricity generation, the expansion of SMRs reduces the need for additional natural gas-fired generation so that by 2035, low-emissions sources account for over 55% of the US data centre electricity supply mix (Figure 2.21). Beyond 2035, the addition of carbon capture to some natural gas-fired power plants is expected to further boost the supply of low-emissions electricity to data centres.

Figure 2.21 ▷ Electricity generation for data centres in the United States and China in the Base Case, 2020-2035



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Natural gas is set to continue to dominate the near-term data centre electricity supply in the United States, with coal predominant in China

In **China**, as data centres are located mostly in the east of the country, their electricity supply is dominated by coal with a share of about 70%, followed by renewables with nearly 20%, nuclear close to 10% and natural gas accounting for the remainder. Between 2024 and 2030 both coal and renewables – mostly solar PV and wind – add about 90 TWh to the data centre electricity supply. The increase in renewables is supported by their rising share in the grid electricity mix, provincial colocation mandates and policies to prioritise the construction of data centres in renewables-rich western China. After 2030, the introduction of SMRs significantly boosts the nuclear share of the data centre electricity mix. Between 2030 and 2035, the rise in renewables and nuclear pushes coal into decline. By 2035, both sources together make up 60% of China's data centre electricity supply (Figure 2.21).

In **Europe**, renewables and nuclear are set to supply most of the additional electricity required, with their combined share rising to 85% by 2030. **Japan** and **Korea** together account for about 5% of global data centre electricity demand today, a share they are expected to retain to 2030. Renewables and nuclear are set to provide nearly 60% of the

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electricity consumed by data centres in 2030, up from 35% today. The rest of the world is responsible for about 10% of total data centre electricity generation, with **Southeast Asia** and **India** accounting for a significant portion of that. In both regions, coal remains a key pillar of the data centre electricity supply, but renewables are projected to eclipse it by 2035.

Box 2.6 ▷ How are utilities in the United States planning to meet additional electricity demand?

Over the course of 2024, many US utilities revised their load growth projections, anticipating a significant increase in electricity demand from data centres, manufacturing and – to a lesser degree – electric vehicles and electric heating. They are seeking to meet this additional demand primarily by building new natural gas-fired power plants and expanding the capacity of low-emissions sources of electricity, most notably wind and solar PV, as well as battery storage to facilitate the integration of variable renewables. In its updated integrated resource plan for North Carolina, Duke Energy, for example, has announced plans to build 7 GW of renewables capacity, 3.6 GW of natural gas-fired capacity, 1.8 GW of pumped storage hydro and 1.1 GW of battery storage until 2035, while Dominion Energy plans to add 21 GW of low-emissions power generation, including 1.3 GW of SMRs, as well as 5.9 GW of gas-fired capacity and 4.5 GW of battery storage across Virginia and North Carolina until 2039.

Integrated resource plans are comprehensive, regularly updated plans that utilities employ to outline their generation requirements over periods ranging from 5 to more than 20 years, identifying the necessary resources to meet anticipated demand and ensure reliable service while balancing economic, environmental and regulatory constraints and objectives. They are essential for planning and are mandated by regulatory authorities in 33 states.

As of Q4 2024, the integrated resource plans of the United States' utilities call for the installation of an additional 260 GW of wind and solar PV capacity until 2035, 20 GW less than planned at the end of last year. Gas-fired capacity is set to grow by 84 GW over the same period, 32 GW higher than planned at the end of 2023 (RMI, 2025). Utilities cite grid constraints and low reserve margins in several systems, as well as the high reliability needs of data centres, as primary reasons for the renewed dash for gas.

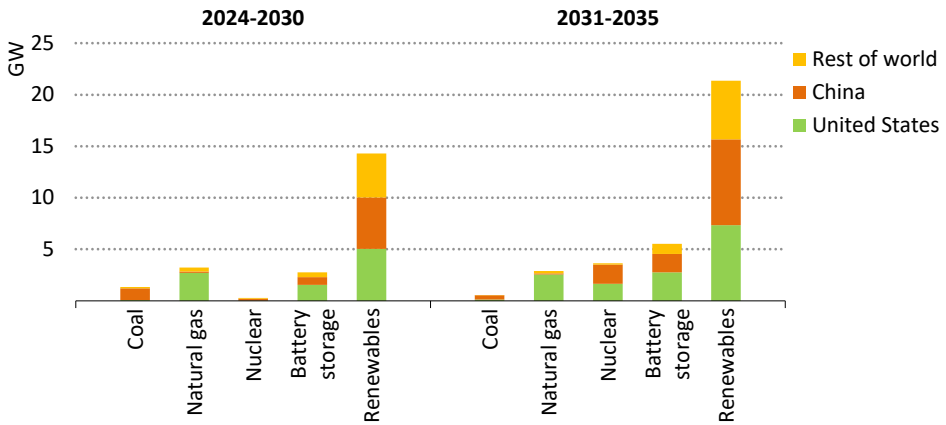
Modernising and expanding the grid to facilitate the integration of variable renewables and ensure reliability is another key feature of many integrated resource plans. Grid modernisation involves upgrading infrastructure, rolling out smart grid technologies and enhancing cybersecurity measures. The goal is to manage electricity flows efficiently and minimise the risk of outages.

In the Base Case, the growth in global data centre electricity consumption sees the installation of over 320 GW of additional electricity generating capacity between 2024 and 2035, including around 45 GW of battery storage, nearly 80% of it in the United States and

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China. Renewables account for nearly two-thirds of this additional capacity. Natural gas-fired capacity also grows, driven primarily by the expansion of natural gas plants to supply data centres in the United States. Gas turbines could also be deployed as a backup power source for large data centres receiving electricity from the grid (Box 2.7). More than half of the additional gas-fired capacity is installed before 2030 to meet immediate electricity needs, while after 2030, growth in nuclear picks up so that, together with renewables, low-emissions sources cover all of the additional demand growth. Nearly 20 GW of new nuclear capacity is commissioned between 2030 and 2035, mostly from SMRs in the United States and China (Figure 2.22).

Figure 2.22 ▷ Annual average data centre power supply capacity additions by fuel and region in the Base Case, 2024-2030 and 2031-2035



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While renewables account for two-thirds of the additional data centre electricity supply capacity, significant volumes of natural gas, coal and nuclear capacity are also added

Box 2.7 ▷ Backup power for data centres

Backup power sources for data centres are critical in ensuring uninterrupted operations during power outages. The primary technologies employed include batteries, diesel generators, gas generators and gas turbines. For additional redundancy, data centre operators also usually request a minimum of two lines to connect their facilities to the electricity grid.

Battery-based uninterruptible power supply systems provide instantaneous power during outages, thereby preventing operational disruptions. They frequently also offer protection against power surges and voltage fluctuations. However, the duration of power supply from these systems is typically limited, ranging from a few minutes to an hour. The system is therefore usually only designed to bridge the time it takes for alternative backup power sources to start up.

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Diesel generators can deliver sustained power for extended outages. Their reliability and ability to handle substantial power loads are crucial for data centre operations. However, diesel generators emit pollutants. Additionally, they generate significant noise and require onsite storage of fuel, which can be a constraint in urban environments.

Gas generators provide a cleaner alternative to diesel, emitting fewer pollutants and generally exhibiting higher fuel efficiency. However, they are dependent on a continuous gas supply, which can be a vulnerability if the supply is disrupted. The initial capital expenditure associated with gas generators is also higher compared with diesel generators.

Gas turbines are another potential option. They offer a reliable and continuous power source, essential for maintaining operations during prolonged outages. Gas turbines are more efficient and less emissions-intensive than diesel generators. However, the initial investment for gas turbines, which tend to be significantly larger than generators, is substantial, and permitting burdens can be more significant. Their size and the need to combine several units to achieve the necessary reliability required of a backup power source make them suitable only for very large data centres. Just like gas generators, they are reliant on an uninterrupted supply of gas. As start-up times are longer than for gas or diesel engines (around one minute for aeroderivative turbines and five minutes for single-shaft, utility-size gas turbines), the battery backup will need to be sized for longer runtimes accordingly.

Each backup power technology must be evaluated according to the specific requirements and constraints of the data centre, especially the specified availability levels. A combination of different technologies can enhance the robustness and reliability of backup systems. Section 2.6.3 looks at the possibility of leveraging backup power systems for flexibility.

2.5.4 Electricity supply in the sensitivity cases

The sensitivity cases examine the uncertainties surrounding future electricity demand from data centres and the implications for electricity generation over the next five to ten years. Across all cases, renewables play a pivotal role in meeting the growing electricity demand. However, fossil fuels remain important for meeting the near-term surge in demand up to 2030.

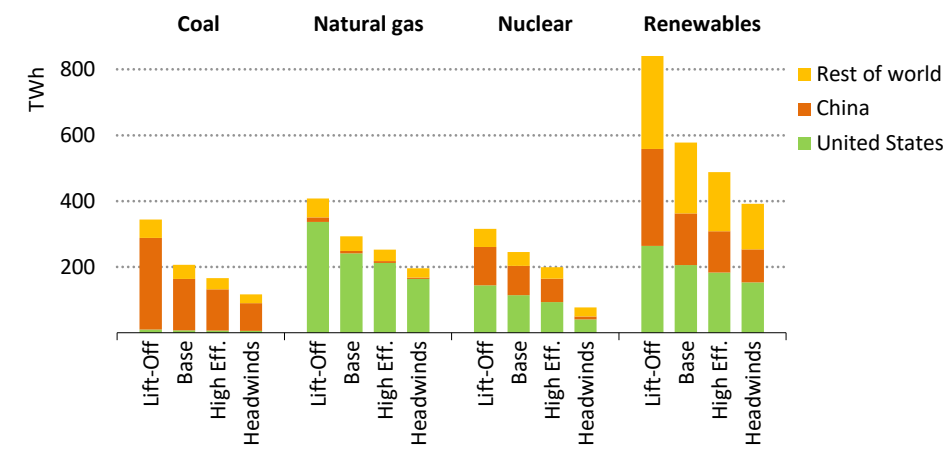
Across all cases, **renewables** meet most of the additional electricity demand from data centres to 2035. In the High Efficiency Case and the Headwinds Case, global data centre-related electricity generation grows more slowly than in the Base Case. In the High Efficiency Case it rises to about 1 100 TWh by 2035, more than 15% lower than in the Base Case. In the Headwinds Case it reaches 790 TWh, more than 40% lower than in the Base Case. In both of these cases, renewables meet 55% or more of the increase in data centre electricity demand to 2035, compared with around 50% in the Base Case, although in both cases, the increase is smaller in absolute terms. In the Lift-Off Case, where global electricity generation

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associated with data centres surges to nearly 2 000 TWh by 2035, 45% higher than in the Base Case, around 45% more renewable electricity generation is added between 2024 and 2035, but long grid connection queues mean that most of the additional increase beyond that is met by fossil fuels.

Across the outlook period, fossil fuels, particularly **coal** and **natural gas**, remain crucial for addressing potential demand spikes (2.24). In the Lift-Off Case, between 2024 and 2030, nearly 50% of the additional electricity generated for data centres comes from fossil fuels. Natural gas-fired power generation grows about 1.5 times faster than in the Base Case, with the United States experiencing the most significant absolute increase. Similarly, coal-fired generation grows twice as fast, with China contributing most of the additional generation. For the period between 2024 and 2035, fossil fuels account for about 35% of the additional electricity consumed by data centres globally. In the High Efficiency Case and the Headwinds Case, fossil fuels respectively supply around 35% and 15% of the additional electricity, as opposed to 28% in the Base Case. The share of fossil fuels in total electricity generation for data centres in 2035 remains at about 40% across all cases.

Figure 2.23 ▷ Electricity generation for data centres by fuel and case, 2035



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In all cases, fossil fuels remain an important element of the data centre electricity supply mix by 2035

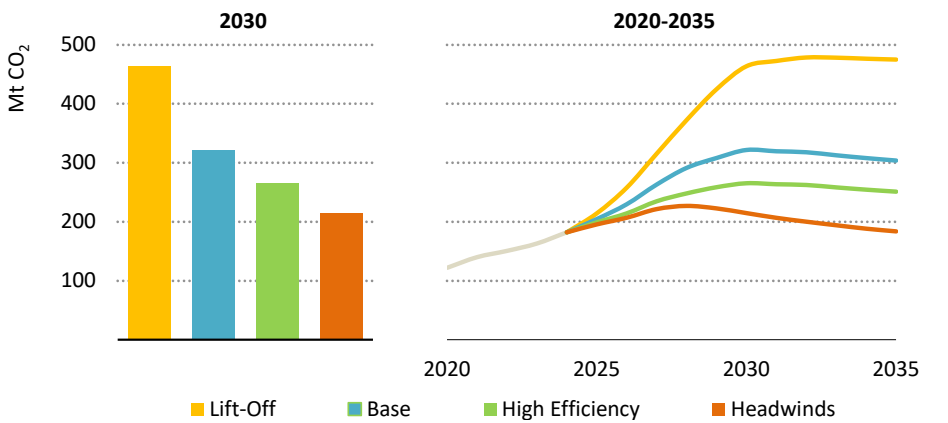
Note: High Eff. = High Efficiency.

The Base, High Efficiency and Lift-Off Cases all see an increase in the contribution of **nuclear** power to the data centre electricity supply between 2030 and 2035, driven mainly by the commissioning of SMRs in the United States and China, which together account for over 80% of total global nuclear electricity generation for data centres. The share of nuclear in the data centre electricity mix ranges between 16% and 18% in the Base, High Efficiency and Lift-Off Cases. It is only in the Headwinds Case, with its less favourable environment for AI and data centre operators, that these investments do not materialise, and nuclear electricity is

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sourced entirely from large-scale reactors connected to the grid, with the nuclear share dropping to about 10% of the data centre electricity supply mix by 2035.

Figure 2.24 ▷ CO_2 emissions associated with electricity generation for data centres by case, 2030 and 2020-2035



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Data centre electricity supply-related CO_2 emissions peak at 215 Mt CO_2 to 320 Mt CO_2 in all cases except the Lift-Off Case, which sees a plateau at around 475 Mt CO_2 in the 2030s

In the Base, High Efficiency and Headwinds Cases, CO_2 emissions from electricity generation for data centres peak around or before 2030. However, in the Lift-Off Case, which sees significantly higher levels of fossil fuel-based electricity generation, they continue to increase until the early 2030s, peaking at nearly 1.5 times the maximum emissions level of the Base Case.

In the Headwinds Case, emissions peak earlier than in the Base Case, dropping to about 215 Mt CO_2 in 2030. This is primarily due to the lower data centre electricity demand growth. In 2030, CO_2 emissions in the High Efficiency Case are around 265 Mt CO_2, nearly 20% lower than in the Base Case and roughly 55% of the CO_2 emissions of the Lift-Off Case.

2.6 Data centre interactions with the electricity grid

2.6.1 Is there a risk of delays in connecting data centres to the grid?

The global expansion of data centre capacity faces risks from grid connection delays, particularly in regions experiencing high concentrations of demand growth. Connection queues for new data centres can already be long in many key regions (Table 2.4). In recent years, several jurisdictions have placed moratoriums on new data centres while system operators process the backlog of connection requests and assess the capacity of the grid to meet additional connections.

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Lessons from the biomedical sector highlight broad themes in AI-driven innovation:

- First, the importance of large, high-quality datasets. AlphaFold's success was only possible due to the availability of the PDB, an open-access, high-quality dataset assembled over 50 years of global collaboration.
- Second, the need for rigorous, human-driven validation before deployment. AI predictions must be rigorously evaluated to ensure they are accurate and reliable. While AI can identify patterns and generate insights at scale, human oversight remains essential to assess causality, interpretability and unintended consequences.
- Third, the challenge of translating AI advances into commercial applications – even when AI accelerates discovery or problem-solving, real-world implementation is often constrained by technical or regulatory barriers.

The breakthrough of AlphaFold has generated a huge amount of interest in AI-driven innovation (it could be described as the "ChatGPT moment" of the field). While it holds important lessons, care needs to be taken in applying the paradigm of one sector to another. As we shall see throughout this chapter, the challenges of energy sector innovation are sometimes characterised by similar extremely complex searches for a "needle in a haystack": a new material, molecule or enzyme, for example. However, often the challenge lies as much, or more, in integrating this new material, catalyst or molecule into highly complex products like batteries; integrating new products into large, complex and slow-moving industrial supply chains; and concurrently addressing myriad enabling conditions, including infrastructure, policy support and consumer preferences.

4.3 How can AI accelerate solutions to energy innovation challenges?

4.3.1 Overview of the innovation cycle

Although innovation pathways are complex, they broadly progress through the following phases:

- Applied research focuses on understanding, measuring and manipulating the fundamental physical, chemical or biological foundation of a technology. This early phase broadly corresponds to Technology Readiness Levels (TRLs) 1 to 4.[1] This phase is sometimes characterised by the painstaking search for promising molecules, materials or chemistries.
- The outcomes of applied research enable the development of prototypes (TRLs 5-6). A key challenge here is the integration of new concepts developed in the laboratory into more complex working devices.

[1] TRLs are a scale from 1 to 9 used for reporting on the level of maturity of new technologies. Originally conceived by the National Aeronautics and Space Administration, they are now widely used as a measure of innovation.

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- Prototypes are then scaled up during the demonstration phase (TRLs 7-8), where full-scale commercial units are operated in real-world conditions. During this phase, scale-up can falter under the weight of rising costs and implementation risks.
- After demonstration, the gap between costs and revenues begins to fall, but new energy technologies may still remain unprofitable. However, purchases by early adopters (TRL 9) can begin to build a durable market.
- As products enter the mainstream, their users and manufacturers continue to see opportunities for improvement, generating both modest design adjustments and additional low-TRL ideas to be tested and scaled up as potential successors.

In this chapter, the stages of innovation are broadly grouped together into two phases: proving (up to TRL 6) and scaling (from TRL 7).

4.3.2 Integrating AI into the innovation process

The full extent of the domains of many scientific fields makes exhaustive experimental searches impractical. There are, for instance, more potential inorganic compounds with four unique atoms than there are people on earth. Historically, researchers have relied on laborious and expensive trial-and-error processes to navigate these vast design spaces. For example, in developing catalytic synthesis processes to make ammonia – now the second-most widely used industrial chemical globally – researchers at BASF spent over three years systematically screening more than 2 500 catalysts.

AI can accelerate the search for candidate molecules, materials or chemistries in a number of ways (see Figure 4.5):

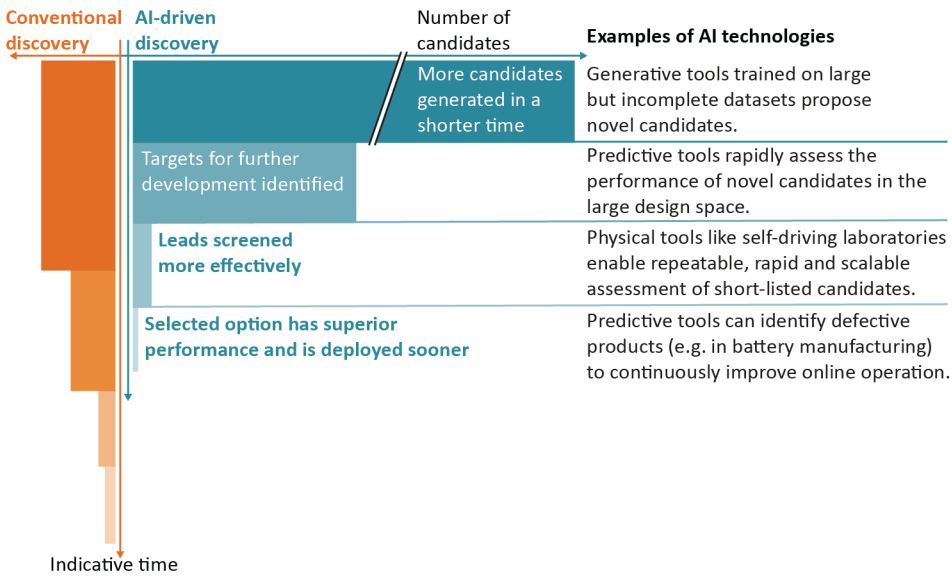
- First, predictive AI models can learn from available experimental data on catalyst designs, perovskite materials or battery chemistries to make predictions about which candidates could meet desired performance characteristics. Examples of this include the use of AI models for protein simulation in drug discovery (see Spotlight above).
- Second, generative AI models can propose novel options (e.g. materials that have never been synthesised) that can be explored and tested both computationally and experimentally. For example, Microsoft's MatterGen diffusion model can propose novel, stable and unique materials with desirable properties when trained on existing databases of relevant materials.
- Finally, large language models can help scientists access and organise vast bodies of academic literature and extract information on existing technologies, approaches and candidate designs.

Once a promising candidate is identified, the AI model's assessment of its characteristics must be validated. This validation can be achieved using high-throughput experimentation (HTE), which also closely resembles the approach taken by the BASF designers of the ammonia synthesis catalysts: once they had narrowed their pool of materials down to a subset of iron-based materials, they developed standardised laboratory-scale reactors and

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conducted parallel tests across multiple candidates, dramatically accelerating the refinement process. Modern HTE is more mechanised and performed on a larger scale but is based on the same fundamental principles.

Figure 4.5 ▷ Conceptual approaches to searching large solution spaces, conventionally and led by AI



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AI-led design approaches can use existing information to systematically expand the search space to consider more options than could ever be experimentally feasible

AI can itself be integrated into this process using self-driving laboratories, which represent the next step in the evolution of HTE. They use iterative decision making to further accelerate the prototyping process. By automating both the execution of experiments and the selection of the next set of candidates to test, self-driving laboratories enable continuous, repeatable iterations that would be impractical with human-led experimentation alone. The physical AI system that guides this process can retrain itself with the outcomes of each experiment, guiding the research towards promising directions. The Canadian firm Telescope Innovation, for example, has combined robotic automation, process analytics and machine learning to demonstrate new production methods for battery materials.

The A-Lab at the US Government's Lawrence Berkeley National Laboratory provides a compelling demonstration of the potential of self-driving laboratories for accelerating materials innovation. This system synthesised 41 materials – initially predicted by the Materials Project, an open-access database of material properties – without prior knowledge of their structure by leveraging a knowledge base of more than 24 000 scientific publications. The entire workflow, from synthesis to characterisation, was automated through robotic

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handling, with the system autonomously adjusting parameters when experiments failed. This iterative approach shortened the feedback loop between hypothesis and characterisation from weeks to days, ensuring that underperforming experiments contributed more effectively to the development cycle (Szymanski, et al., 2023). Although very promising, these tools have yet to reach the stage of complete autonomy, and human researchers are still required for comprehensive characterisation, and purity and defect control (Peplow, 2024).

Despite their transformative potential, several significant barriers limit the widespread uptake of self-driving laboratories in energy innovation. The most immediate challenge is cost – developing a self-driving laboratory requires substantial investment in both robotics and machine learning infrastructure, with costs potentially reaching tens of millions of dollars. Given the field's nascent state, these systems remain largely bespoke, further increasing the complexity and expense of implementation. The quality and design of experiments – determining which variables to test and how to structure the exploration of the solution space – remain heavily dependent on human expertise and intuition.

4.3.3 What energy technology areas will be accelerated by AI?

Despite the substantial opportunities for AI to accelerate energy innovation, its impact across different fields of scientific research will vary. Technology areas most suited to high-impact applications of AI in the innovation process include:

- Diverse solution spaces that have high levels of combinatorial complexity that cannot be explored by trial-and-error experimentation but for which a large number of potential candidates are well-described in the training data. Catalyst design and pharmaceutical sectors are thus obvious candidates for AI-powered innovation because of the permutational complexity of different atomic combinations.
- Structured and high-quality data for building effective AI models. For example, perovskite materials hold promise for solar PV applications, with over 10 million possible perovskite structures. However, only about 1 000 have been synthesised, limiting the training data available for AI models. While machine learning can estimate properties of unexplored perovskites, its reliability remains constrained by the low availability of high-quality, real-world data.
- Straightforward testing and verification. Training datasets are, by their nature, incomplete and often exclude important metrics, such as energy efficiency and manufacturing costs. AI-proposed reverse osmosis membranes, for instance, can be tested using standard seawater in laboratory conditions; by comparison, plastics recycling is complicated by the wide variety of potential input materials, which are difficult to replicate prior to deployment.
- A receptive market and infrastructure environment that requires neither changes in regulation nor behaviour and does not need investment in new assets in adjacent sectors. Applying AI to technology problems that are likely to yield "drop-in" solutions could produce faster impacts than applying AI to energy sector challenges that face more complicated market conditions

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Table 4.1 applies these four criteria to key energy technology areas, showing specific examples of innovations that could be transformative within those areas, and highlighting the extent to which the innovation challenges correspond to the criteria. Many energy technology challenges have a high degree of complexity, but many also lack adequate data for AI to process that complexity. As with the scale-up of new energy technologies today, almost all products of AI-driven innovation will face some hurdles in incorporation into existing, interconnected energy systems.

Table 4.1 ▷ Illustrative assessment of the potential for AI to accelerate progress against selected key energy technology challenges

Technology challenge	Solution space complexity	Structured data availability	Pre-deployment verification	Integration and scaling
Synthetic fuels - Catalysts with high efficiency, selectivity and stability	High	High	High	Medium
Hydrogen electrolysis - Low-cost, highly efficient and durable electrolyser catalysts	High	High	High	Medium
Carbon capture, utilisation and storage - Stable CO_2 capture materials with high affinity and low energy penalty	High	Medium	High	Medium
Electric vehicles - Novel battery chemistries using cheap materials (e.g. sodium-ion, solid-state)	High	Medium	High	Medium
High-temperature heat storage - Stable phase change materials with high conductivity and latent heat	High	Low	High	Medium
Desalination - Productive, stable and energy efficient reverse osmosis membranes	High	Medium	High	Low
Advanced biofuels - Improved performance of enzymes and yeasts for 2nd/3rd generation biofuels	High	High	Medium	Low
Solar photovoltaics - Efficient, stable, scalable perovskite cells without critical mineral inputs	High	Low	High	Medium
High-temperature heat pumps - Identification of working fluids which phase change at high temperatures	Medium	Low	High	Medium
Long-duration energy storage - Cheaper, efficient redox-flow or other long-duration batteries	High	Low	Medium	Medium
Decarbonised cement - Cement production from calcium silicate raw materials	Medium	Low	Medium	Medium
Plastics recycling - Energy-efficient upgrading of pyrolysis oils	Medium	Medium	Low	Medium
Effective nuclear fusion - Fusion reaction control	High	Low	Low	Low

High Medium Low

Note: Green indicates a high degree of alignment between the criteria and the technology challenge, suggesting AI is more likely to have meaningful impact in the sector; orange indicates some alignment, and that innovation in the sector could benefit from AI; red indicates low alignment, suggesting a possible hurdle to AI deployment.

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As people and businesses in emerging market and developing economies remain open to adopting the latest available tools in their daily lives and processes, the prospect of leapfrogging older technologies remains alluring. Just as the mobile phone revolution bypassed landline expansion, AI-driven applications may enable some of these economies to sidestep legacy systems and adopt cutting-edge energy management solutions directly. Governments can encourage this by integrating AI objectives into national energy strategies, promoting local AI skills development through partnerships with universities, training centres and research centres, and offering clear incentives for private-sector involvement. Blended finance mechanisms – combining concessional loans, guarantees and private capital – could also help mitigate risks for projects aiming to build both digital and energy infrastructure.

Ultimately, each country requires customised solutions that reflect its unique mix of resources, markets and regulatory contexts. For some, attracting data centres through reliable green power could be a catalyst for modernisation; for others, the priority might be smaller-scale digital tools that bolster rural electrification or reduce transmission losses. In all cases, addressing both energy and digital connectivity gaps together is crucial. By fostering local data collection, developing talent and creating robust policy frameworks, emerging market and developing economies can harness AI to drive more inclusive, future-proof growth – growth that integrates renewable energy expansion, meets rapidly rising demand and supports new industries in the process.

5.7 The AI and energy policy landscape

The growing role of AI in the global economy in recent years has led to an evolving policy landscape. As of 2025, most economies have adopted a national AI strategy. Recently adopted strategies have focused notably on creating frameworks to foster AI development and use in national economies, including the energy sector. This section looks at national policy frameworks and their impacts on the AI–energy nexus.

5.7.1 The enabling role of government in AI development

National AI strategies often involve government financial support for the emergence of an AI industry. As it stands, support generally focuses on three main components: the development of AI models and use through research and development (R&D) programmes, direct support for data centre development, and domestic incentives for manufacturing chips and semiconductors. For example, Japan invested USD 13 billion in 2023 alone to build the foundations of a semiconductor and AI-related technology sector (Japan, Ministry of Finance, 2024) part of its pledge to invest JPY 10 trillion (USD 65 billion) by 2030 (The Japan Times, 2024).

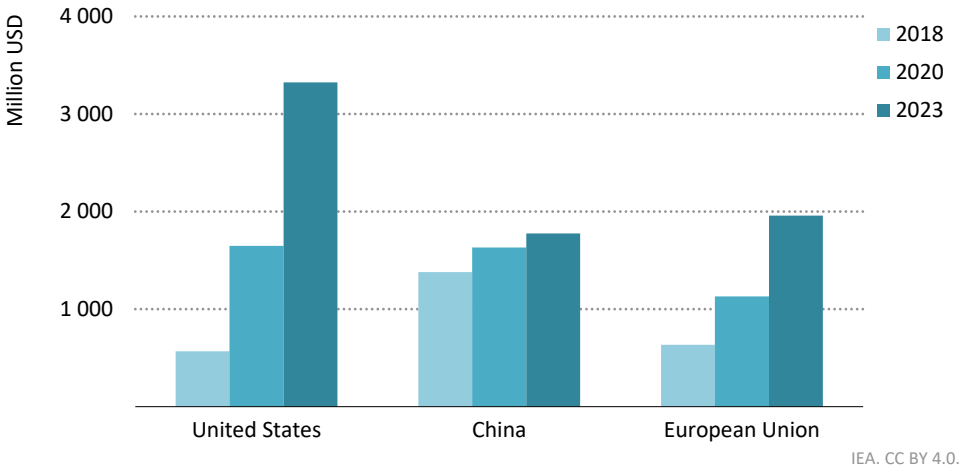
R&D has grown sharply in recent years, with close to USD 7 billion of disbursement from governments for AI-related R&D projects in 2023, close to three times the amount spent in 2018. This growth can be attributed to three main regions: the United States, the European Union and China (see Chapter 4). However, future plans indicate a broadening of

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this trend. Brazil recently published its Artificial Intelligence Plan 2024-2028, with a key objective to boost its R&D in this area by earmarking USD 4.6 billion in the next four years (Brazil, Ministry of Science, Technology and Innovation, 2024). Egypt also released its Second National AI Strategy in 2025, with the objective of creating a national AI fund equivalent to USD 430 million to USD 860 million (Egypt, The National Council for Artificial Intelligence, 2025).

In 2024, India facilitated the creation of three AI Centres of Excellence worth USD 120 million (India, Ministry of Electronics and IT, 2025) and launched the IndiaAI Mission and its Semicon India programme, with an initial budget of USD 9.2 billion to accelerate the uptake of AI infrastructure and necessary components such as microchips (India, Ministry of Electronics & IT, 2024).

Figure 5.26 ▷ Government R&D for AI-related projects in digital software and innovation



R&D surged in all regions in the past five years, from USD 2.6 billion in 2018 to more than USD 7.0 billion in 2023

Notes: R&D = research and development. Government R&D includes primary and secondary AI projects.

Sources: IEA analysis based on the respective government websites and Archaya and Arnold (2019), and China Central Government (n.d.).

Korea's Restriction of Special Tax Treatment Act incentivises data centre development through tax credits of up to 12% of facility investment costs for AI and cloud companies (Ministry of Economy and Finance of Korea, 2023). China also provides significant tax breaks with its High and New-Technology Enterprise status, which involves reduced corporate income tax rates from 25% to 15% (The State Council of the People's Republic of China, 2019). Thailand's Board of Investment offers substantial long-term tax incentives, up to 13 years of tax exemption from machinery to raw material import duties (Thailand Board of Investment, 2025).

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5.7.2 Energy and AI policy frameworks

As noted above and in Chapter 2, data centres bring both opportunities and challenges for the grid. Policy frameworks have started being developed to incentivise data centre development outside areas of grid congestion (W Media, 2023a). The Korean Government offers a 50% discount on the electricity facility levy to data centres built outside the Seoul metropolitan area (W Media, 2023b). Beyond incentives, some jurisdictions impose stricter rules on data centre expansion, for example through energy performance requirements or a moratorium in some cases. In 2018, Beijing banned the construction or expansion of data centres in the city, except for cloud computing data centres with a power usage effectiveness (PUE) of 1.5 or less (The People's Government of Beijing Municipality, 2018). The Netherlands and Singapore imposed a moratorium on data centres in 2019 as they were reconsidering their data centre strategies with the influx of data centres – both moratoriums have since been lifted (Data Centre Dynamics, 2022; The Straits Times, 2022). In South Africa, the National Policy on Data and Cloud has designated locations for data centres to reduce stress on the national grid (Republic of South Africa, 2024). The Electric Reliability Council of Texas recently enabled a Large Load Revision Request Package requiring certain information about all loads to ensure a more rational load queue and encourage flexibility (The National Law Review, 2025).

Several countries have mandated minimum energy performance standards. These performance standards specifically focus on the PUE, or the ratio between the power consumption of the whole facility against the consumption of the IT equipment. The National Australian Built Environment Rating System, in place since January 2025, is the first and only mandatory labelling programme for data centres, ranging between a PUE of 1.07 (6 stars – market leading) to 2.42 (1 star – making a start). Requirements can be downscaled to IT-specific equipment like data storage, network equipment and servers (NABERS, 2024). The 2019 EU regulation on ecodesign requirements for servers and data storage products imposes both power efficiency requirements (gradually increasing between March 2020 and January 2023) and material efficiency requirements for data storage devices, memory and processors. Germany expanded the scope of the energy reuse factor, which only accounts for reused heat and energy, requiring facilities to reach 10% in 2026 and 15% by 2028 (Germany, Federal Ministry for Economic Affairs and Climate Action, 2023).

Table 5.1 ▷ Data centre energy efficiency mandates for selected economies

Region	PUE (2023)	PUE mandate
Australia	1.44	1.4 by 2025
China	1.56	1.5 by 2025
France	1.36	40% building energy use reduction by 2030
Germany	1.42	1.2 by 2026 (new), 1.3 by 2030 (existing)
Japan	1.53	1.4 by 2022
California (United States)	1.21	1.5 by 2014
Global	**1.43**	-

Note: PUE = power usage effectiveness.

Sources: Based on government websites and Masanet (2024).

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Table 5.2 ▷ Policy landscape in selected economies

Economy	National strategy	Government financial support			Reporting requirements		Performance mandates	
		R&D	Data centres	Chips	Emissions	Electricity consumption	PUE	WUE
Argentina	●							
Australia	●	●	●		●	●	●	
Brazil	●	●	●	●				
Canada	●	●	●	●	○	○		
China	●	●	●	●		●	●	●
European Union	●	●	●	●	●	●		
France	●	●		●	●		●	
Germany	●	●		●	●	●	●	
India	●	●	○	●				
Indonesia	●	●	●	●				
Italy	●	●		●	●			
Japan	●	●	●	●	●	●	●	
Korea	●	●	●	●				
Mexico	●							
Russian Federation	●	●	●		●			
Saudi Arabia	●	●	●	●				
South Africa	●	●						
Türkiye	●	●	●	●	●			
United Kingdom	●	●	●	●	●	●		
United States	●	●	●	●	○	○	○	

Notes: ○ = subnational only; PUE = power usage effectiveness; WUE = water usage effectiveness.

Reporting requirements for data centres are on a voluntary basis in most jurisdictions as of 2025, but some mandatory schemes are being developed (see Table 5.1 and 5.2). The EU Corporate Sustainability Reporting Directive notably entered into force in 2023 and requires direct and indirect greenhouse gas emissions reporting from large and listed companies, that is, including emissions from electricity consumption or data provider subsidiaries. The Energy Efficiency Directive sets annual reporting obligations across 31 metrics for data centre owners and operators, and the implementation of certified energy management systems, such as ISO 50001, for large energy users, replacing the previous four-year audit requirement. The Delegated Regulation 2024/1364 set reporting requirements for data centres specifically on PUE and a water usage effectiveness (WUE) metric (Box 5.4 contains a discussion on water use by data centres). Singapore reformed its Green Data Centre Standard in 2020 to meet the ISO 50001 standard for energy management. Canada's Energy Star voluntary programme provides its own certificate for each piece of data centre equipment, including data storage, large network equipment, servers and uninterruptible power supplies.

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Hardware efficiency is key, yet software efficiency is discussed only in a handful of countries. Mitigating the rise in data centre energy demand could focus on fostering the efficiency of AI services themselves. This can take various forms, from smaller models requiring fewer parameters to optimised AI model training. In June 2024, France published benchmarks for measuring and reducing the environmental impact of AI, with 26 recommendations and best practices for the conception and development of AI models.

Box 5.4 ▷ Water use by data centres: How thirsty is AI?

Data centres require large amounts of water – both directly for cooling onsite as well as indirectly for water consumption associated with semiconductor manufacturing and energy supply. Water use varies significantly by data centre, depending on the cooling technology, the local climate and the source of electricity supply. For example, direct expansion cooling (which accounts for around four-fifths of cooling in enterprise data centres) is many times less water intensive than airside economiser and adiabatic cooling with water-cooled chillers (used in about half of hyperscale data centres).[2] Based on estimates of the current breakdown of cooling technologies, we estimate that on average a 100 MW hyperscale data centre in the United States consumes around 2 million litres per day in total – equivalent to about 6500 households – with over 60% of this being indirect water use.

We estimate that global water consumption for data centres is currently around 560 billion litres per year, and this could rise to around 1200 billion litres per year in 2030 in the Base Case (Figure 5.27). Global water withdrawals[3] for data centres show a similar steep increase to 2030. About two-thirds of the consumption in 2023 was associated with primary energy supply and electricity generation, a further one-quarter with direct cooling and the remainder for water used in semiconductor and microchip manufacturing.

A number of factors determine overall water intensity, and this changes the relative water demands associated with direct and indirect operations over time. For energy supply, water withdrawals depend heavily on the mix of technologies used for electricity generation, with solar PV and wind using one-hundredth of the water that fossil sources use, or less (IEA, 2016). This means that the water use associated with energy supply is growing more slowly than data centre electricity demand, as more electricity is being generated from renewable sources. Conversely, at the manufacturing stage, almost 90% of water consumption is associated with ultra-pure water production, needed to produce

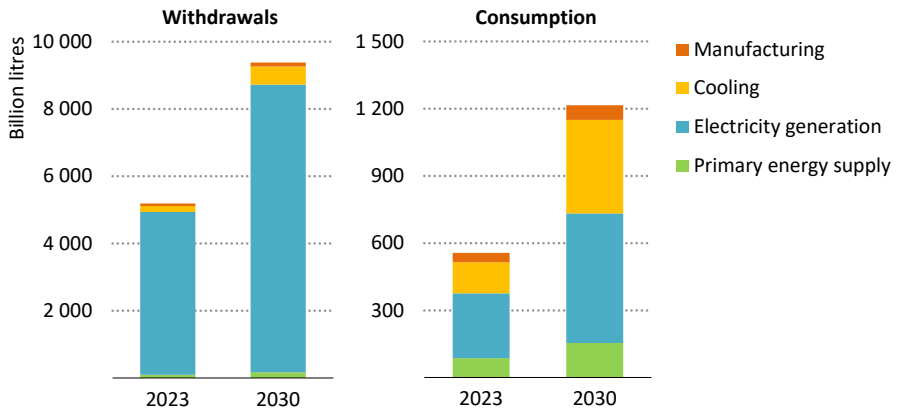
[2] In direct expansion cooling, refrigerant circulates directly through indoor coils to cool the data centre air. In airside economiser and adiabatic cooling with water-cooled chiller cooling, water evaporates directly to provide cooling supplemented by water-cooled chiller systems, which use cool water drawn from natural resources or produced by cooling towers.

[3] Withdrawals are the total amount of water withdrawn from sources including surface water and groundwater. Consumption represents the portion of withdrawals not returned to the original water source after use but lost, e.g. through evaporation.

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microchips from semiconductors. In the Base Case, water consumption from chip manufacturing for data centres grows more than 50% from 2023 levels to around 70 billion litres in 2030 – faster than new servers are added to data centres – driven by an increasing number of accelerated servers, which tend to contain more chips than conventional servers.[4]

Figure 5.27 ▷ Water withdrawals and consumption by data centres in the Base Case, 2023 and 2030



Water consumption more than doubles between 2023 and 2030

Notes: Assumes a fixed water use per area of wafer area between 2023 and 2030. Assumes direct WUE by cooling technologies equivalent to 2023. Water requirements are quantified for "source-to-carrier" primary energy production (oil, gas, coal and hydrogen), a definition which includes production, processing and transport. Water withdrawals and consumption for bioenergy account for water use for processing. For electricity generation, freshwater requirements are for the operational phase, including cleaning, cooling and other process-related needs. Electricity generation includes fossil fuels, nuclear, modern bioenergy and renewables waste, solar PV, concentrating solar power, wind and geothermal.

Sources: IEA modelling and analysis based on Harris, et al. (2019), Hamed et al. (2022), IEA (2016), Lei, et al. (2025), Lei and Masanet (2022), and Shehabi, et al. (2024).

In some countries, such as the United States, withdrawals for data centres today equate to less than 10% of annual municipal water withdrawals, but elsewhere, the water demands of data centres could compete with water for agricultural irrigation and municipal uses and even impact the supply chains that underpin microchip manufacturing. In Chinese Taipei, for example, semiconductor manufacturers were subject to water restrictions during a drought in 2021, requiring water use reduction strategies to be implemented. Around half of the consumption by 2030 is in Asia Pacific

[4] While recycling efforts and water efficiency measures could reduce water use, there is also evidence that advanced chip manufacturing has higher than average water intensity, and large chip manufacturers have reported increased water use per wafer since 2020.

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countries, where a typically warm and humid climate makes cooling more water intensive. For instance, Microsoft estimated that the water use effectiveness (WUE)[5] for direct cooling in Asia Pacific was 1.65 litres per kilowatt hour (kWh) – more than three times the global average of its data centres of around 0.5 litres per kWh (Microsoft, 2022). In Europe, the Climate Neutral Data Centre Pact, signed by 97 operators and associations, targets a reduction in WUE to less than 0.4 litres per kWh by 2040 (Climate Neutral Data Center, 2023).

The number of people exposed to water stress is set to increase by at least 50% by 2050 with climate change (Munia, et al., 2020). Siting new data centres in areas of low water stress is a straightforward way to ensure sustainability ambitions are met, but innovation could also help quench the water needs of data centres and ensure that data centres are not adding to water stress risks in a warming climate. For example, direct liquid cooling (where liquid coolants circulate directly through servers with a coupling to a heat exchanger) and immersion cooling (where servers are submerged in a non-conducting dielectric fluid) can reduce direct water consumption significantly (Kong, et al., 2024). Operators including AWS and Google have pledged to be "water positive" by 2030, by combining recycling and replenishment programs with reductions in the direct WUE of their operations. (Google, 2024; Amazon, 2023).

Semiconductor manufacturing facilities are also making headway. By adjusting the design of rinse tanks, ultra-pure water use can be reduced and, increasingly, manufacturers are installing onsite water recycling technologies. AI can also play a role in improving WUE by dynamically adjusting cooling requirements based on real-time data and predictive algorithms. AI has applications in desalination technologies (see Chapter 4), which could help expand the supply of usable water for cooling in coastal regions. AI could also help address water stress risks in the economy as a whole via improved resource management, such as predictive analytics for leak detection, smart irrigation and the optimisation of water infrastructure.

5.8 An exploratory approach to determine the potential impact of AI on emissions

The world is on track to witness a global temperature rise of 2.4 °C by 2100 under a trajectory determined by prevailing policy settings as of October 2024. This is substantially higher than the Paris Agreement goal to pursue efforts to limit the global temperature rise to 1.5 °C

[5] WUE is the ratio of total direct water use to IT electricity consumption. The 2024 United States Data Center Energy Usage Report estimates an average WUE value of 0.36 litres per kWh in the United States for 2023; several data centre operators have reported significantly higher values in the range of 1-1.5 litres per kWh for their global operations (Shehabi, et al., 2024; Equinix, 2023; Google, 2024). Disclosure of WUE in company reports is less common than for similar sustainability metrics relating to energy use or greenhouse gas emissions, increasing the uncertainty in current and projected water use by the industry as a whole.

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Methodology and data tables

Methodology for data centre energy demand

The modelling of data centre electricity demand relies on a bottom-up approach developed by the Lawrence Berkeley National Laboratory over the past two decades. In this modelling approach, IT equipment shipments are the key driver of data centre electricity demand. We analyse three types of IT equipment: servers, storage systems, and network equipment[1]. The last category refers to network equipment hosted within data centre facilities to connect servers and storage devices to the data network. It should not be confused with the data transmission network, which connects data centres and end-users (for example, 5G network towers). The latter falls outside the scope of the modelling of data centre electricity consumption in this study.

The central input to the model is the annual shipment of servers. These come from:

- IDC's (International Data Corporation), which provides shipment projections for the period 2019-2028 (IDC, 2024)
- These are triangulated with additional data inputs from Omdia (OMDIA, 2025), SemiAnalysis (SemiAnalysis, 2025), and Borderstep Institute (Hintemann, Hinterholtzer, and Konrat, 2024), and additional literature (Koomey, 2007), (Koomey, 2011), (Shehabi, et al., 2024), (Shehabi, 2018), (Shehabi, et al., 2016), (Gartner, 2014a, 2014b, 2014c, 2015a, 2015b, 2015c, 2015d, 2016a, 2016b, 2017a, 2017b, 2017c, 2018a, 2018b, 2020), (Masanet, et al., 2020), (Malmodin, et al., 2024).

The stock of storage drives is derived from hard-disk drive shipment data from Forbes (Forbes, 2021) and the split between HDDs and solid-state drives from (SSDs) (Shehabi, et al., 2024). The stock of network equipment is estimated based on server port density.

We distinguish three types of data centres, which serve as archetypes in this model: enterprise data centres, colocation and service provider data centres, and hyperscale data centres.

The technical characteristics of the server stock, such as lifetime and power consumption, as well as operational characteristics like idle power ratio and utilisation rates, are based on estimates from the United States (Shehabi, et al., 2024). Similarly, for storage drives, the split between storage technologies and average utilisation rates is also based on US estimates. The characteristics of storage drives are assumed to be constant across all data centre types. The network port distribution is also assumed to be constant, with one exception: specific InfinityBand-like network equipment, whose stock depends solely on accelerated servers.

Based on these datasets and input assumptions, we estimate the installed capacity for each type of IT equipment. It is important to note that these values differ from the maximum

[1] See definitions in section 2.1.

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designed capacity, as they consider only the installed units of each IT equipment type and do not reflect total rack capacity.

The regional allocation of global installed IT capacities relies on several factors. The primary driver is the regional breakdown provided by our third-party data provider (IDC), which is based on market dynamics in each region. To achieve finer regional granularity, we also consider the level of digitalisation of economies based on the digital adoption index (World Bank, 2016) and the development of the local data centre market, using publicly available data on data centre market revenues (Statista, 2024), (Turner&Townsend, 2024).

Network equipment is assumed to have a 100% utilisation rate. Storage systems utilisation rates are considered constant. Idle power assumptions are based on trends observed in the SERT database for conventional servers and estimates from the literature for accelerated servers (SPEC, 2024), (Shehabi, et al., 2024).

Aggregation of utilisation rates is conducted by data centre type. The equation for the electricity consumption of servers is as follows:

$$E = (P_{max} - P_{idle}) * u + P_{idle}$$

Where:

- P_{max} is the maximum power draw of an operating server (distinct from the maximum rated power, especially for accelerated servers).
- P_{idle} is the power drawn by a server when not processing useful tasks.
- u is the server utilisation rate.

For each region and data centre type, IT electricity demand is multiplied by the corresponding Power Usage Effectiveness (PUE) to obtain the total electricity demand of the infrastructure and hosted IT equipment.

PUE primarily accounts for cooling equipment, power supply equipment, and lighting. Power supply equipment and lighting are collectively referred to as “auxiliary equipment”. Data centre type influences PUE due to variations in infrastructure efficiency, climate also affects PUE by directly impacting cooling requirements. PUE estimates are based on regional climate and data centre type (enterprise, colocation and service provider, and hyperscale) (Lei and Masanet, 2022). We assume that regional differences within the same data centre category arise from variations in cooling needs. The relative evolution of PUE over time is informed by improvements reported in company-level data (Google, 2025).

The simplified equation for data centre electricity demand in each region is as follows:

$$E_{data\ centre} = \sum_{i\ =\ data\ centre\ type} (E_{server,i} + E_{storage,i} + E_{network,i}) * PUE_i$$

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Data tables

General note to the tables

This annex includes the following datasets:

- **Table A.1 - World Data centres by case:** Includes global historical and projected data by case and data centre type (hyperscale, colocation and service provider and enterprise) for the following metrics:
  - Total and IT installed capacity (GW)
  - Power usage effectiveness
  - Load factor (%)
  - Total and IT electricity consumption (TWh)
- **Table A.2 - Data centres installed capacity by region:** Includes regional historical and projected total and IT installed capacity (GW) for the Base Case
- **Table A.3: Data centres power usage effectiveness and load factor by region**
- **Table A.4: Data centres electricity consumption by region**

Tables A.2 A.3 and A.4 include data for these regions: world, North America, United States, Central and South America, Europe, Africa, Middle East, Asia Pacific and China. The definitions for regions are in Annex B.

Both in the text of this report and in these annex tables, rounding may lead to minor differences between totals and the sum of their individual components.

Annex A licencing

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Table A.1: World data centres by case

				Base		Lift-Off		High Efficiency		Headwinds	
	2020	2023	2024	2030	2035*	2030	2035*	2030	2035*	2030	2035*
Installed capacity (GW)											
Total	60	83	97	226	277	305	404	185	221	158	160
Hyperscale	20	31	36	85	103	108	139	89	103	62	64
Colocation and service provider	19	27	35	86	116	118	172	93	115	59	66
Enterprise	20	25	27	55	58	78	93	3	3	36	31
IT	38	57	68	174	228	233	330	153	196	122	132
Hyperscale	17	27	31	77	94	98	127	81	95	56	58
Colocation and service provider	11	17	23	65	96	89	142	70	100	44	54
Enterprise	10	13	14	32	38	46	61	2	2	21	20
Power usage effectiveness											
Total	1.53	1.43	1.41	1.29	1.21	1.30	1.22	1.21	1.13	1.28	1.20
Hyperscale	1.19	1.15	1.14	1.10	1.09	1.10	1.09	1.10	1.09	1.10	1.09
Colocation and service provider	1.67	1.56	1.53	1.33	1.21	1.33	1.21	1.32	1.15	1.33	1.21
Enterprise	2.05	1.95	1.92	1.71	1.54	1.72	1.54	1.67	1.46	1.71	1.53
Load factor (%)											
Total	51	49	49	48	49	47	49	49	50	48	50
Hyperscale	56	54	53	51	52	50	51	51	52	51	53
Colocation and service provider	50	48	48	47	48	47	48	47	49	48	50
Enterprise	48	46	45	45	46	44	46	45	47	45	48
Electricity consumption (TWh)											
Total	269	361	416	946	1 193	1 264	1 719	792	972	669	707
Hyperscale	100	148	166	378	466	479	626	397	472	279	293
Colocation and service provider	85	112	144	355	493	482	721	385	490	246	285
Enterprise	85	100	106	213	234	303	372	10	10	144	128
IT	176	252	295	733	985	972	1 409	657	864	522	587
Hyperscale	84	129	146	342	427	434	574	360	432	253	269
Colocation and service provider	51	72	94	266	406	361	594	291	425	185	235
Enterprise	42	51	55	124	153	176	242	6	7	84	84

*2035 numbers serve as exploratory scenarios given the high level of uncertainty around data centre demand growth.

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Table A.2: Data centres installed capacity by region

	2020	2023	2024	Base Case 2030
Total installed capacity (GW)				
World	60	83	97	226
North America	24	35	43	102
United States	23	35	42	100
Central and South America	0.3	0.4	0.4	0.8
Europe	13	15	16	27
Africa	0.3	0.3	0.4	0.7
Middle East	0.3	0.3	0.4	0.7
Asia Pacific	21	30	36	92
China	14	20	24	67
IT installed capacity (GW)				
World	38	57	68	174
North America	17	26	32	82
United States	17	26	31	81
Central and South America	0.2	0.2	0.2	0.5
Europe	8	10	11	21
Africa	0.1	0.2	0.2	0.5
Middle East	0.1	0.2	0.2	0.4
Asia Pacific	12	19	24	67
China	8	13	16	49

Table A.3: Data centres power usage effectiveness and load factor by region

	2020	2023	2024	Base Case 2030
Power usage effectiveness				
World	1.53	1.43	1.41	1.29
North America	1.39	1.32	1.32	1.24
United States	1.39	1.31	1.32	1.23
Central and South America	1.82	1.73	1.70	1.50
Europe	1.57	1.47	1.45	1.29
Africa	1.97	1.85	1.81	1.59
Middle East	2.06	1.96	1.92	1.70
Asia Pacific	1.68	1.55	1.50	1.35
China	1.67	1.56	1.50	1.35
Load factor (%)				
World	51	49	49	48
North America	53	51	50	48
United States	53	51	50	49
Central and South America	50	48	47	46
Europe	51	49	48	48
Africa	49	46	46	45
Middle East	49	46	46	45
Asia Pacific	50	48	48	47
China	50	48	48	47

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Table A.4: Data centres electricity consumption by region

	2020	2023	2024	Base Case 2030
Total electricity consumption (TWh)				
World	269	361	416	946
North America	112	158	187	434
United States	108	154	183	426
Central and South America	1.5	1.5	1.7	3.3
Europe	57	66	68	113
Africa	1.1	1.3	1.4	2.9
Middle East	1.1	1.3	1.5	3.0
Asia Pacific	93	128	150	378
China	62	84	102	277
IT electricity consumption (TWh)				
World	176	252	295	733
North America	80	120	142	351
United States	78	117	139	345
Central and South America	0.8	0.8	1.0	2.2
Europe	36	45	47	88
Africa	0.6	0.7	0.8	1.8
Middle East	0.5	0.7	0.8	1.7
Asia Pacific	55	82	100	281
China	37	54	68	205

IEA. CC BY NC SA 4.0.

Chapter 2: Energy for AI

ACER (European Union Agency for the Cooperation of Energy Regulators) (2024), Electricity infrastructure development to support a competitive and sustainable energy system 2024 Monitoring Report, https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER_2024_Monitoring_Electricity_Infrastructure.pdf

American Electric Power (2025), Indiana Michigan Power Receives Order in Large Load Settlement, https://www.aep.com/news/stories/view/10037/

Bloomberg (2024), AI Needs So Much power, It's Making Yours Worse, https://www.bloomberg.com/graphics/2024-ai-power-home-appliances/

BNEF (Bloomberg New Energy Finance) (2025), Renewable Energy Project Database, https://www.bnef.com/assets/power

BNEF (2021), Data Centers and Decarbonization, https://www.eaton.com/content/dam/eaton/company/news-insights/energy-transition/documents/bnef-eaton-statkraft-data-center-study-en-us.pdf

Bundesnetzagentur (Federal Network Agency - Germany) (2025), Monitoring report 2024, https://www.bundesnetzagentur.de/EN/Areas/Energy/DataCollection_Monitoring/start.html

Cambridge Centre for Alternative Finance (2025), Cambridge Bitcoin Electricity Consumption Index, https://ccaf.io/cbnsi/cbeci

Chen, J. and Ran, X. (2019), Deep Learning With Edge Computing: A Review, Proceedings of the IEEE, vol. 107(8), pp. 1655–1674, https://doi.org/10.1109/JPROC.2019.2921977

Cisco (2008), Cisco Visual Networking Index Cisco Visual Networking Index Forecast 2007-2012, Retrieved from https://www.cisco.com/c/dam/en_us/about/ac78/docs/IPTrafficStudy_current_speakerseries_APAC.pdf

Cisco (2015), Global - 2015 Year in Review, https://www.cisco.com/c/dam/m/en_us/solutions/service-provider/vni-forecast-highlights/pdf/Global_2015_Year_in_Review.pdf

Cisco (2019), Cisco VNI Forecast update, https://www.ieee802.org/3/ad_hoc/bwa2/public/calls/19_0624/nowell_bwa_01_190624.pdf

CRS (Congressional Research Service) (2023), Electricity Transmission: What is the role of Federal Government?, https://crsreports.congress.gov/product/pdf/R/R47862

Delli Abo, M. (2024), An Efficiency Comparison of NPU, CPU, and GPU When Executing an Object Detection Model YOLOv5, https://kth.diva-portal.org/smash/record.jsf?pid=diva2%3A1886212&dswid=9457

Deloitte (2024), Powering artificial intelligence - A study of AI's environmental footprint - today and tomorrow, https://www.deloitte.com/global/en/issues/climate/powering-ai.html

IEA. CC BY 4.0.

ENTSO-E (European Network of Transmission System Operators for Electricity) (2022), Overview of Transmission Tariffs in Europe: Synthesis 2020, https://eepublicdownloads.entsoe.eu/clean-documents/mc-documents/l_entso-e_TTO-Report_2020_03.pdf#:~:text=Sweden%20Deep%20Generators%20or%20consumers,First%20connection%20charges

EPRI (Electric Power Research Institute) (2024), Utility Experiences and Trends Regarding Data Centers: 2024 Survey, https://www.epri.com/research/products/000000003002030643

Ericsson (2024), Impact of GenAI on mobile network traffic, https://www.ericsson.com/4acd55/assets/local/reports-papers/mobility-report/documents/2024/emr-november-2024-genai-article.pdf

Gartner (2024), Gartner Predicts Power Shortages Will Restrict 40% of AI Data Centers By 2027, https://www.gartner.com/en/newsroom/press-releases/2024-11-12-gartner-predicts-power-shortages-will-restrict-40-percent-of-ai-data-centers-by-20270

Goldman Sachs (2024), Generational growth, AI/data center's global power surge and the Sustainability impact, https://www.goldmansachs.com/insights/goldman-sachs-research/gs-sustain-generational-growth-ai-data-centers-global-power

Grid Strategies (2024), 2023 Transmission Congestion Report, https://gridstrategiesllc.com/wp-content/uploads/Grid-Strategies_2023-Transmission-Congestion-Report.pdf

GSMA (2025), Rethinking Mobile Phones: The Business Case for Circularity, https://www.gsma.com/solutions-and-impact/connectivity-for-good/external-affairs/climate/rethinking-mobile-phones/

GSMA (2024), Mobile Net Zero 2024: State of the Industry on Climate Action, https://www.gsma.com/solutions-and-impact/connectivity-for-good/external-affairs/wp-content/uploads/2024/02/Mobile-Net-Zero-2024-State-of-the-Industry-on-Climate-Action.pdf

IDC (International Data Corporation) (2024a), Datacenter Trends: Sustainable Builds and CO_2 Emissions, https://www.idc.com/getdoc.jsp?containerId=IDC_P33186

IDC (2024b),The Future of Next-Gen AI Smartphones, https://blogs.idc.com/2024/02/19/the-future-of-next-gen-ai-smartphones/

IEA (International Energy Agency) (2025), Electricity 2025, https://www.iea.org/reports/electricity-2025

IEA (2024), World Energy Outlook 2024, https://www.iea.org/reports/world-energy-outlook-2024

IEA (2023), Data Centres and Data Transmission Networks, https://www.iea.org/energy-system/buildings/data-centres-and-data-transmission-networks

IEA. CC BY 4.0.

ITU (2025), DataHub, https://datahub.itu.int/query/

ITU (2018), L.1450: Methodologies for the assessment of the environmental impact of the information and communication technology sector, https://www.itu.int/rec/T-REC-L.1450-201809-I/en

JLARC (Joint Legislative Audit and Review Commission) (2024), Data Centers in Virginia, https://jlarc.virginia.gov/landing-2024-data-centers-in-virginia.asp

Kaack, L. H. et al. (2022), Aligning artificial intelligence with climate change mitigation, Nature Climate Change, vo. 12(6), pp. 518–527, https://doi.org/10.1038/s41558-022-01377-7

Kamiya, G., & Coroamă, V. C. (2025), Data Centre Energy Use: Critical Review of Models and Results, https://www.iea-4e.org/edna/publications/

Kelly, C. et al (2016), Balancing Power Systems With Datacenters Using a Virtual Interconnector, https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=7452537

Koomey, J. G. (2007), Estimating total power consumption by servers in the U.S. and the world, http://uploadi.www.ris.org/editor/1203418697svrpwrusecompletefinal.pdf

Leiserson, C. E. et al. (2020), There's plenty of room at the Top: What will drive computer performance after Moore's law?, Science, vol. 368, https://doi.org/10.1126/science.aam9744

Li, Y. et al. (2024), The Unseen AI Disruptions for Power Grids: LLM-Induced Transients, https://doi.org/10.48550/arXiv.2409.11416

Luers, A. et al. (2024), Will AI accelerate or delay the race to net-zero emissions?, Nature, vol. 628, pp. 718-720, https://doi.org/10.1038/d41586-024-01137-x

Magnum Economics (2024), The Impact of Data Centers on Virginia's State and Local Economies 5th Biennial Report, Northern Virginia Technology Council, https://info.nvtc.org/acton/attachment/45522/f-1c3915e6-b8b1-4914-818e-9fae14877a3d/1/-/-/-/-/2024%20NVTC%20Data%20Center%20Report.pdf

Malmodin, J. and Lundén, D. (2018), The energy and carbon footprint of the global ICT and E&M sectors 2010–2015, Sustainability, https://doi.org/10.3390/su10093027

Malmodin, J. et al. (2024), ICT sector electricity consumption and greenhouse gas emissions – 2020 outcome, Telecommunications Policy, vol. 48, https://doi.org/10.1016/j.telpol.2023.102701

Masanet, E., Lei, N. and Koomey, J. (2024), To better understand AI's growing energy use, analysts need a data revolution, Joule, 8(9), pp. 2427-2436, https://doi.org/10.1016/j.joule.2024.07.018

Meltwater (2024), Digital 2025 Global Overview Report, https://www.meltwater.com/en/global-digital-trends

IEA. CC BY 4.0.

Mytton D. et al. (2023), Stretched grid? Managing data center energy demand and grid capacity, Oxford Open Energy, vol. 2, https://doi.org/10.1093/ooenergy/oiad014

Mytton, D., Lundén, D. and Malmodin, J. (2024), Network energy use not directly proportional to data volume: The power model approach for more reliable network energy consumption calculations. Journal of Industrial Ecology, vol. 28(4), pp. 966-980, https://doi.org/10.1111/jiec.13512

NERC (North American Electric Reliability Corporation) (2025), Incident Review Considering Simultaneous Voltage-Sensitive Load Reductions, https://www.nerc.com/pa/rrm/ea/Documents/Incident_Review_Large_Load_Loss.pdf

NESO (National Energy System Operator – United kingdom) (n.d.), Daily Balancing Services Use of System (BSUoS) Cost Data, https://www.neso.energy/data-portal/daily-balancing-costs-balancing-services-use-system

Nicholas Institute for Energy, Environment & Sustainability (2025), Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems, https://nicholasinstitute.duke.edu/sites/default/files/publications/rethinking-load-growth.pdf

OMDIA (2025), Data Center Building and Investment Intelligence Service, https://omdia.tech.informa.com/advance-your-business/cloud-and-data-center/data-center-building-and-investment-intelligence-service

OpenAI (2025), OpenAI Roadmap and characters, https://community.openai.com/t/openai-roadmap-and-characters/1119160

Qiu, X. et al. (2021), Can Federated Learning Save The Planet?, https://doi.org/10.48550/arXiv.2010.06537

RMI (Rocky Mountain Institute) (2025), The State of Utility Planning, 2024 Q4, https://rmi.org/the-state-of-utility-planning-2024-q4/

Rouphael, R. B. et al. (2023), The Impact of Networks in the Greenhouse Gas Emissions of a Major European CSP, https://doi.org/10.1109/ICECET58911.2023.10389498

Schneider Electric (2024), Artificial Intelligence and Electricity: A System Dynamics Approach, https://www.se.com/ww/en/insights/sustainability/sustainability-research-institute/artificial-intelligence-electricity-system-dynamics-approach/

Semianalysis (2025), Custom Market Intelligence, https://semianalysis.com/datacenter-industry-model/

SemiAnalysis (2024), AI Datacenter Energy Dilemma - Race for AI Datacenter Space // Gigawatt Dreams and Matroyshka Brains Limited By Datacenters Not Chips, https://semianalysis.com/2024/03/13/ai-datacenter-energy-dilemma-race/

Shehabi, A. et al. (2024), 2024 United States Data Center Energy Usage Report, https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf

IEA. CC BY 4.0.

SPEC (2024), SPECpower_ssj2008 Results, https://www.spec.org/power_ssj2008/results/power_ssj2008/

Tan, T. and Cao, G. (2023), Deep Learning on Mobile Devices With Neural Processing Units, Computer, vol. 56(8), pp. 48–57, https://doi.org/10.1109/MC.2022.3215780

TenneT (n.d.) TenneT publiceert de Annual Market Update 2023 [TenneT publishes the Annual Market Update 2023], https://www.tennet.eu/nl/nieuws/tennet-publiceert-de-annual-market-update-2023

Utility Dive (2024), Can regulators protect small customers from rising transmission costs for big data centers?, https://www.utilitydive.com/news/regulators-protect-small-customers-rising-transmission-costs-data-centers/735155/

Weinbach, M. and Bajarin, B. (2024), The NPU Wattage Advantage, https://creativestrategies.com/research/white-paper-the-npu-wattage-advantage/

World Bank (2024a), Individuals using the Internet (% of population) (database) accessed February 2025, https://data.worldbank.org/indicator/IT.NET.USER.ZS

World Bank (2024b). Measuring the Emissions & Energy Footprint of the ICT Sector: Implications for Climate Action, http://documents.worldbank.org/curated/en/099121223165540890

Chapter 3: AI for energy optimization

5GLOGINNOV (2024), Using 5G technologies to innovate logistics and ports for a sustainable future, https://5g-loginnov.eu/

Abdelalim, M. A. et al. (2025), An Analysis of Factors Contributing to Cost Overruns in the Global Construction Industry, Buildings, vol. 15(1)(18), https://www.mdpi.com/2075-5309/15/1/18

Abdullayeva, F. and Imamverdiyev, Y. (2019), Development of Oil Production Forecasting Method based on Deep Learning, Statistics, Optimization, and Information Computing, https://www.semanticscholar.org/paper/Development-of-Oil-Production-Forecasting-Method-on-Abdullayeva-Imamverdiyev/4558ab6a3c1abb70e12d8c3bc7a3d84af08af1e2

ACER (European Union Agency for the Cooperation of Energy Regulators) (n.d.), ACER-FSR report launch: benefit sharing to promote more efficient investments in energy infrastructure, https://www.acer.europa.eu/public-events/acer-fsr-report-launch-promote-more-efficient-investments-energy-infrastructure

AEMO (n.d.), International system operator collaboration, https://aemo.com.au/initiatives/major-programs/international-system-operator-collaboration

AER (Australian Energy Regulator) (n.d.), https://www.aer.gov.au/system/files/4.%20AER%20explanatory%20statement%20-%20efficiency%20benefit%20sharing%20scheme%20-%20November%202013.pdf

IEA. CC BY 4.0.

Chapter 4: AI for energy innovation

Adarsh, D. et al. (2022), Autonomous optimization of non-aqueous Li-ion battery electrolytes via robotic experimentation and machine learning coupling, Nature, vol. 13, https://www.nature.com/articles/s41467-022-32938-1

AIonics (2024), Aionics Unveils Expansion into Aviation Batteries Fueled by Strategic Investments and Industry Partnerships, https://www.globenewswire.com/news-release/2024/12/05/2992366/0/en/Aionics-Unveils-Expansion-into-Aviation-Batteries-Fueled-by-Strategic-Investments-and-Industry-Partnerships.html

An, K. et al. (2023), A comprehensive review on regeneration strategies for direct air capture, Journal of CO_2 utilization, vol. 76, https://doi.org/10.1016/j.jcou.2023.102587

Annevelink, E. et al. (2022), AutoMat: Automated Materials Discovery, https://arxiv.org/pdf/2011.04426

Attia, P. et al. (2020), Closed-loop optimization of fast-charging protocols for batteries with machine learning, Nature, pp. 397-402, https://doi.org/10.1038/s41586-020-1994-5

Cao, R. et al. (2025), Model-constrained deep learning for online fault diagnosis in Li-ion batteries over stochastic conditions, Nature, https://doi.org/10.1038/s41467-025-56832-8

CATL (2025), Smart Manufacturing, https://www.catl.com/en/manufacture/

Chen, C. et al. (2024), Accelerating Computational Materials Discovery with Machine Learning and Cloud High-Performance Computing: from Large-Scale Screening to Experimental Validation, Journal of the American Chemical Society, https://pubs.acs.org/doi/abs/10.1021/jacs.4c03849

El-Bousiydy, H. et al. (2021), What Can Text Mining Tell Us About Lithium-Ion Battery Researchers' Habits?, https://doi.org/10.1002/batt.202000288

Friedman, J. (2024), Artificial Intelligence for Climate Change Mitigation Roadmap, https://www.energypolicy.columbia.edu/publications/icef-2024-roadmap-on-icef-artificial-intelligence-for-climate-change-mitigation-roadmap-second-edition/

Frith, J., Lacey, M. and Ulissi, U. (2023), A non-academic perspective on the future of lithium-based batteries, Nature, vol. 14, https://doi.org/10.1038/s41467-023-35933-2

Haowei, H. et al. (2023), EVBattery: A Large-Scale Electric Vehicle Dataset for Battery Health and Capacity Estimation. Cornell University, https://arxiv.org/abs/2201.12358

Huang, Q. et al. (2023), A review of the application of artificial intelligence to nuclear reactors: Where we are and what's next, Heliyon, https://doi.org/10.1016/j.heliyon.2023.e13883

IBM (International Business Machines Corporation) (2025), Accelerated Discovery of Battery Materials, https://research.ibm.com/projects/accelerated-discovery-of-battery-materials

IEA. CC BY 4.0.

IEA (International Energy Agency) (2025), The Path to a New Era for Nuclear Energy, https://www.iea.org/reports/the-path-to-a-new-era-for-nuclear-energy

IEA (2024a), The Future of Geothermal Energy, https://www.iea.org/reports/the-future-of-geothermal-energy

IEA (2024b), EV Life Cycle Assessment Calculator, https://www.iea.org/data-and-statistics/data-tools/ev-life-cycle-assessment-calculator

IEA (2020), Clean Energy Innovation, https://www.iea.org/reports/clean-energy-innovation

Jie, B. et al. (2019), Machine Learning Coupled Multi-Scale Modeling for Redox Flow Batteries, Advanced Theory and Simulations, https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/adts.201900167

Kang, Y. and Kim, J. (2024), ChatMOF: an artificial intelligence system for predicting and generating metal-organic frameworks using large language models, Nature, https://doi.org/10.1038/s41467-024-48998-4

Liu, Z., Yu, A. and Lee, J. (1999), Synthesis and characterization of LiNi1−x−yCoxMnyO2 as the cathode materials of secondary lithium batteries, Journal of Power Sources, pp. 416-419, https://doi.org/10.1016/S0378-7753(99)00221-9

Magdău, I. et al. (2023), Machine learning force fields for molecular liquids: Ethylene Carbonate/Ethyl Methyl Carbonate binary solvent, Nature, https://www.nature.com/articles/s41524-023-01100-w

Microsoft (2024), Accelerating the discovery of battery materials with AI, Science, https://www.science.org/content/article/ai-driven-collaboration-rapidly-identifies-new-battery-material

Milne, R., John, J. and Novik, M. (2024), How Europe's battery champion descended into crisis, FT, https://www.ft.com/content/1e7ab9cc-bedd-4c59-bd02-31530e7c1002

Open Source Battery Data (2025), https://github.com/lappemic/open-source-battery-data

Padhi, A., Nanjundaswamy, K. and Goodenough, J. (1997), Phospho‐olivines as Positive‐Electrode Materials for Rechargeable Lithium Batteries, Journal of The Electrochemical Society, vol. 1188, https://ui.adsabs.harvard.edu/link_gateway/1997JElS..144.1188P/doi:10.1149/1.1837571

Park, H. et al. (2024), A generative artificial intelligence framework based on a molecular diffusion model for the design of metal-organic frameworks for carbon capture, Communications Chemistry, https://doi.org/10.1038/s42004-023-01090-2

Peplow, M. (2024), Robot chemist sparks row with claim it created new materials, Nature, https://doi.org/10.1038/d41586-023-03956-w

Rahmanian, F. et al. (2024), Attention towards chemistry agnostic and explainable battery lifetime prediction, https://doi.org/10.1038/s41524-024-01286-7

IEA. CC BY 4.0.

Ruifeng, T. et al. (2025), BatteryLife: A Comprehensive Dataset and Benchmark for Battery Life Prediction, Cornell University, https://arxiv.org/abs/2502.18807

Sainju, R. et al. (2022), DefectTrack: a deep learning-based multi-object tracking algorithm for quantitative defect analysis of in-situ TEM videos in real-time, Scientific Reports, https://doi.org/10.1038/s41598-022-19697-1

SES AI (2025a), SES AI Unveils an AI-Enhanced 2170 Cylindrical Cell for Humanoid Robotics and Drone Applications at CES 2025, https://investors.ses.ai/news/news-details/2025/SES-AI-Unveils-an-AI-Enhanced-2170-Cylindrical-Cell-for-Humanoid-Robotics-and-Drone-Applications-at-CES-2025/default.aspx

SES AI (2025b), SES AI signs contracts totaling up to $10 million to develop AI-enhanced Li-Metal and Li-ion batteries for EVs with two automotive OEM partners, https://investors.ses.ai/news/news-details/2025/SES-AI-Unveils-an-AI-Enhanced-2170-Cylindrical-Cell-for-Humanoid-Robotics-and-Drone-Applications-at-CES-2025/default.aspx

Shengyu, T. et al. (2025), PulseBat: A field-accessible dataset for second-life battery diagnostics from realistic histories using multidimensional rapid pulse test, Cornell University, https://arxiv.org/abs/2502.16848

Sobes, V. et al. (2021), AI-based design of a nuclear reactor core, Nature, https://doi.org/10.1038/s41598-021-98037-1

Song, Z., et al. (2025), Inverse design of promising electrocatalysts for CO_2 reduction via generative models and bird swarm algorithm, Nature, https://doi.org/10.1038/s41467-024-55613-z

Sriram, A. et al. (2024), The Open DAC 2023 Dataset and Challenges for Sorbent Discovery in Direct Air Capture, ACS Central Science, pp. 923-941, https://doi.org/10.1021/acscentsci.3c01629

Szymanski, N. et al. (2023), An autonomous laboratory for the accelerated synthesis of novel materials, Nature, https://doi.org/10.1038/s41586-023-06734-w

Tao, S. et al. (2023), Collaborative and privacy-preserving retired battery sorting for profitable direct recycling via federated machine learning, Nature, https://doi.org/10.1038/s41467-023-43883-y

The Chemical Engineer (2024), AI put to work in push for rapid battery development, https://www.thechemicalengineer.com/news/ai-put-to-work-in-push-for-rapid-battery-development/

White, A. et al. (2024), Alarming structural error rates in MOF databases used in data driven workflows identified via a novel metal oxidation state-based method, Chemrxiv, https://chemrxiv.org/engage/chemrxiv/article-details/6706b96312ff75c3a1fb0365

Winter, M., Barnett, B., Xu, K. (2018), Before Li Ion Batteries, https://doi.org/10.1021/acs.chemrev.8b00422

IEA. CC BY 4.0.

Wright, A. et al. (2024), Transitioning metal–organic frameworks from the laboratory to market through applied research, Nature Materials, pp. 178-187, https://doi.org/10.1038/s41563-024-01947-4

Yao, N. et al. (2022), Applying Classical, Ab Initio, and Machine-Learning Molecular Dynamics Simulations to the Liquid Electrolyte for Rechargeable Batteries, Chemical Reviews, 10970-11021, https://doi.org/10.1021/acs.chemrev.1c00904

Chapter 5: Emerging themes on energy and AI

Amazon (2023), Water Stewardship, https://sustainability.aboutamazon.com/natural-resources/water

American Electric Power (2024), I&M and Stakeholders File Large Load Settlement to Advance Grid Reliability and Support Economic Growth, https://www.aep.com/news/stories/view/9883/IM-and-Stakeholders-File-Large-Load-Settlement-to-Advance-Grid-Reliability-and-Support-Economic-Growth-/

Apple (2025), 2024 Environmental Progress Report, https://www.apple.com/environment/

Archaya, A. and Arnold, Z. (2019), Chinese Public AI R&D Spending: Provisional Findings, https://doi.org/10.51593/20190031

Batouta, K., Aouhassi, S. and Mansouri, K. (2024), Energy saving potential in steam systems: A techno-economic analysis of a recycling pulp and paper mill industry in Morocco, Scientific African, http://dx.doi.org/10.1016/j.sciaf.2024.e02375

Bloomberg (2025), Price Tag for Amazon's Mississippi Data Centers Jump to $16 billion, https://www.bloomberg.com/news/articles/2025-01-31/amazon-mississippi-data-center-costs-jump-to-16-billion

BNEF (Bloomberg New Energy Finance) (2025), Renewable Energy Transactions, https://www.bnef.com/interactive-datasets/2d5d59acd9000028?tab=Transactions%20-%20All%20Deals&view=c0cce78b-1e63-4e16-a087-5a788cec9471

Brazil, Ministry of Science, Technology and Innovation (2024), Plano Brasileiro de Inteligência Artificial (PBIA) 2024-2028 [Brazilian Artificial Intelligence Plan (PBIA) 2024-2028], https://www.gov.br/lncc/pt-br/assuntos/noticias/ultimas-noticias-1/plano-brasileiro-de-inteligencia-artificial-pbia-2024-2028

Bremer, C. et al. (2023), Assessing Energy and Climate Effects of Digitalization: Methodological Challenges and Key Recommendations. Network for the Digital Economy and the Environment, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4459526

CDR.fyi (2025), Leaderboards (database) accessed February 2025, https://www.cdr.fyi/leaderboards

Centre for Transport Studies (2015), Explaining “peak car” with economic variables, https://www.transportportal.se/swopec/CTS2016-13.pdf

IEA. CC BY 4.0.

Checkpoint (2025), Cyber Security Report, http://checkpoint.com/security-report/?flz-category=items&flz-item=report--cyber-security-report-2025

China Central Government (n.d.), China Statistical Year Book, https://www.stats.gov.cn/english/Statisticaldata/yearbook/

Climate Neutral Data Center (2023), Climate Neutral Data Center Working Groups, https://www.climateneutraldatacentre.net/working-groups/#water

Covenant of Mayors (2023), Heat recovery from data centres, https://eu-mayors.ec.europa.eu/sites/default/files/2023-10/2023_CoMo_CaseStudy_Stockholm_EN.pdf

Data Centre Dynamics (2022), Dutch government halts hyperscale data centers, pending new rules, https://www.datacenterdynamics.com/en/news/dutch-government-halts-hyperscale-data-centers-pending-new-rules

Digital Refining (2024), Optimising furnace run length in a steam cracker using AI, https://www.digitalrefining.com/article/1003092/optimising-furnace-run-length-in-a-steam-cracker-using-ai

Egypt, The National Council for Artificial Intelligence (2025), Egypt National Artificial Intelligence Strategy Second Edition (2025-2030), https://ai.gov.eg/SynchedFiles/en/Resources/AIstrategy%20English%2016-1-2025-1.pdf

Equinix (2023), Sustainability Report FY2023, https://sustainability.equinix.com/wp-content/uploads/2024/07/Equinix-Inc_2023-Sustainability-Report.pdf

Eurostat (2025), Artificial intelligence by size class of enterprise, https://ec.europa.eu/eurostat/databrowser/product/page/isoc_eb_ai__custom_15180193

Fagnant, D. and Kockelman, K. (2014), The travel and environmental implications of shared autonomous vehicles, using agent-based model scenarios, Transportation Research Part C: Emerging Technologies, https://doi.org/10.1016/j.trc.2013.12.001

Financial Times (2025), Big Tech lines up over $300bn in AI spending for 2025, https://www.ft.com/content/634b7ec5-10c3-44d3-ae49-2a5b9ad566fa

Financial Times (2024), How the US can overcome China’s gallium ban, https://www.ft.com/content/85c875f7-f7c7-41d6-b635-fd7a51074a34

Fitch Ratings (2024a), Fitch Affirms Indonesia's Perusahaan Listrik Negara at 'BBB'; Outlook Stable, https://www.fitchratings.com/research/corporate-finance/fitch-affirms-indonesia-perusahaan-listrik-negara-at-bbb-outlook-stable-16-09-2024/dodd-frank-disclosure

Fitch Ratings (2024b), Eskom Holdings SOC Ltd., https://www.fitchratings.com/entity/eskom-holdings-soc-ltd-80464056

Fitch Ratings (2019), Fitch Affirms Enel Brasil and Its Subsidiaries' Ratings; Outlook Stable, https://www.fitchratings.com/research/corporate-finance/fitch-affirms-enel-brasil-its-subsidiaries-ratings-outlook-stable-18-09-2019

IEA. CC BY 4.0.

Fridstrøm, L. and Østli , V. (2021), Direct and cross price elasticities of demand for gasoline, diesel, hybrid and battery electric cars: the case of Norway, European Transport Research Review, vol. 13, https://doi.org/10.1186/s12544-020-00454-2

Germany, Federal Ministry for Economic Affairs and Climate Action (2023), Bundestag adopts Energy Efficiency Act, https://www.bmwk.de/Redaktion/EN/Pressemitteilungen/2023/09/20230921-bundestag-adopts-energy-efficiency-act-a-clear-legal-framework-for-energy-efficiency.html

Google (2025), Adversarial Misuse of Generative AI, https://cloud.google.com/blog/topics/threat-intelligence/adversarial-misuse-generative-ai

Google (2024), Environmental Report 2024, https://www.gstatic.com/gumdrop/sustainability/google-2024-environmental-report.pdf

GPO-AI (2024), Global Public Opinion on AI https://srinstitute.utoronto.ca/public-opinion-ai

IEA (International Energy Agency) (2025a), Growing geopolitical tensions underscore the need for stronger action on critical minerals security, https://www.iea.org/commentaries/growing-geopolitical-tensions-underscore-the-need-for-stronger-action-on-critical-minerals-security

IEA (2025b), The Path to a New Era for Nuclear Energy, https://www.iea.org/reports/the-path-to-a-new-era-for-nuclear-energy

IEA (2024a), Global Critical Minerals Outlook 2024, https://www.iea.org/reports/global-critical-minerals-outlook-2024

IEA (2024b), Reducing the Cost of Capital, https://www.iea.org/reports/reducing-the-cost-of-capital

IEA (2024c), World Energy Employment, https://www.iea.org/reports/world-energy-employment-2024

IEA (2023a), Cybersecurity – is the power system lagging behind?, https://www.iea.org/commentaries/cybersecurity-is-the-power-system-lagging-behind

IEA (2023b), Electricity Grids and Secure Energy Transitions, https://www.iea.org/reports/electricity-grids-and-secure-energy-transitions

IEA (2021a), Enhancing cyber resilience in electricity systems, https://www.iea.org/reports/enhancing-cyber-resilience-in-electricity-systems

IEA (2021b), Colonial pipeline outage in the United States underscores risks to energy supplies, https://www.iea.org/commentaries/colonial-pipeline-outage-in-the-united-states-underscores-risks-to-energy-supplies

IEA (2020), Power Systems in Transition, https://www.iea.org/reports/power-systems-in-transition

IEA (2016), World Energy Outlook, https://www.iea.org/reports/world-energy-outlook-2016

IEA. CC BY 4.0.

India, Ministry of Electronics & IT (2025), India's AI Revolution, https://pib.gov.in/PressReleasePage.aspx?PRID=2108810

India, Ministry of Electronics & IT (2024), Cabinet Approves Ambitious IndiaAI Mission to Strengthen the AI Innovation Ecosystem, https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2012357

India Today (2021), Maharashtra cyber cell submits report on Mumbai power outage, confirms malware attack hit power grid, https://www.indiatoday.in/india/story/maharashtra-cyber-cell-mumbai-power-outrage-1774522-2021-03-01

Infosys (2024), The Energy Agenda - Partner Content: An AI-first Approach to a Cleaner Greener Mega Campus, https://www.ons.no/article/partner-content-an-ai-first-approach-to-a-cleaner-greener-mega-campus

Japan, Ministry of Finance (2024), 令和７年度経済産業省関連予算のポイント（概要）[Key Points of the FY2025 METI-related Budget (Summary)], https://www.mof.go.jp/policy/budget/budger_workflow/budget/fy2025/seifuan2025/08.pdf

Kamiya, G. and Coroamă, V.C. (2025), Data Centre Energy Use: Critical Review of Models and Results, https://www.iea-4e.org/edna/publications/

Kong, R. et al. (2024), Enhancing data center cooling efficiency and ability: A comprehensive review of direct liquid cooling technologies, https://www.sciencedirect.com/science/article/pii/S0360544224026203

Lei, N. and Masanet, E. (2022), Climate- and technology-specific PUE and WUE estimations for U.S. data centers using a hybrid statistical and thermodynamics-based approach, https://www.sciencedirect.com/science/article/pii/S0921344922001719

Lei, N. et al. (2025), The Water Footprint of Data Center Workloads: A Review of Key Determinants, https://doi.org/10.21203/rs.3.rs-4159702/v1

Levels.fyi (2025), (database) accessed February 2025, https://www.levels.fyi/

Lightcast (2024), (database) accessed January 2025, https://lightcast.io/products/data/overview

LinkedIn (2025), Technical Note - LinkedIn Methodology, https://economicgraph.linkedin.com/content/dam/me/economicgraph/en-us/PDF/ai-data-partnerships-methodology.pdf

Luccioni, A., Strubell, E. and Crawford, K. (2025), From Efficiency Gains to Rebound Effects: The Problem of Jevons' Paradox in AI's Polarized Environmental Debate, https://arxiv.org/abs/2501.16548v1

Masanet, E. (2024), Review and complication of best available data and modeling approaches for assessing the infrastructure energy use of data centers and artificial intelligence in energy demand models.

Microsoft (2024), 2024 Environmental Sustainability Report - Data Fact Sheet, https://www.microsoft.com/en-us/corporate-responsibility/sustainability/report

IEA. CC BY 4.0.

Microsoft (2022), How Microsoft measures datacenter water and energy use to improve Azure Cloud sustainability, https://azure.microsoft.com/en-us/blog/how-microsoft-measures-datacenter-water-and-energy-use-to-improve-azure-cloud-sustainability/

Ministry of Economy and Finance of Korea (2023), **법인세 공제감면** [Corporate tax deduction and reduction], https://www.nts.go.kr/nts/cm/cntnts/cntntsView.do?cntntsId=7987&mi=6561

Mumbower, S., Garrow, L. and Higg, M. (2014), Estimating flight-level price elasticities using online airline data: A first step toward integrating pricing, demand, and revenue optimization, Transportation Research Part A: Policy and Practice, https://doi.org/10.1016/j.tra.2014.05.003

Munia, H. A. et al. (2020), Future Transboundary Water Stress and Its Drivers Under Climate Change: A Global Study, https://doi.org/10.1029/2019EF001321

NABERS (2024), https://www.nabers.gov.au/

Quantum Commodity Intelligence (2025), Carbon (database) accessed February 2025, https://www.qcintel.com/carbon/data/

Republic of South Africa (2024), National Policy on Data and Cloud, https://www.datalaw.africa/wp-content/uploads/2024/06/South-Africas-National-Cloud-and-Data-Policy-20240531.pdf

SemiAnalysis (2025), Datacenter Anatomy Part 2 – Cooling Systems, https://semianalysis.com/2025/02/13/datacenter-anatomy-part-2-cooling-systems/

SemiAnalysis (2023), AI Server Cost Analysis – Memory Is The Biggest Loser, https://semianalysis.com/2023/05/29/ai-server-cost-analysis-memory-is/

Shehabi, A. et al. (2024), 2024 United States Data Center Energy Usage Report, https://eta.lbl.gov/publications/2024-lbnl-data-center-energy-usage-report

Thailand Board of Investment (2025), Investment Promotion Guide, https://www.boi.go.th/upload/content/BOI_A_Guide_EN.pdf

The Japan Times (2024), Japan unveils ¥10 trillion plan to aid domestic chip industry, https://www.japantimes.co.jp/business/2024/11/11/tech/chip-industry-plan/

The National Law Review (2025), Texas Senate Bill 6 and Impacts on Large Load Development in ERCOT, https://natlawreview.com/article/texas-senate-bill-6-and-impacts-large-load-development-ercot

The People's Government of Beijing Municipality (2018), **北京市人民政府**办公厅关于印发市发展改革委等部门制定的《北京市新增产业的禁止和限制目录(2022年版)》**的通知** [Notice of the General Office of the Beijing Municipal People's Government on Printing and Distributing the "Catalogue of Prohibitions and Restrictions on New Industries in Beijing (2022 Edition)" formulated by the Municipal Development and Reform Commission and other departments], https://www.beijing.gov.cn/zhengce/zhengcefagui/201905/t20190522_61514.html

IEA. CC BY 4.0.

The State Council of the People's Republic of China (2019), 关于高新技术企业的所得税优惠政策了解一下 [Learn more about the preferential income tax policies for high-tech enterprises], https://www.gov.cn/fuwu/2019-11/15/content_5452259.htm

The Straits Times (2022), Singapore pilots sustainable way to grow data centre capacity, https://www.straitstimes.com/tech/singapore-pilots-new-scheme-to-grow-data-centre-capacity-with-green-targets

Turner & Townsend (2024), Data centre cost index 2024, https://www.turnerandtownsend.com/insights/data-centre-cost-index-2024/

Uptime Institute (2023), Heat Reuse: A Management Primer, https://uptimeinstitute.com/resources/research-and-reports/heat-reuse-a-management-primer

US EIA (United States Energy Information Administration) (2023), Table 11.1 Reliability Metrics of U.S. Distribution System, https://www.eia.gov/electricity/annual/html/epa_11_01.html

W Media (2023a), Japan Plans to Subsidize Data Center Construction in Hokkaido and Kyushu, https://w.media/japan-plans-to-subsidize-data-center-construction-in-hokkaido-and-kyushu/

W Media (2023b), Korea Provides Incentives to Data Centers Built in Non-Capital Areas, https://w.media/south-korean-govt-provides-incentives-to-data-centers-being-built-in-non-capital-areas/

World Bank (2024a), World Development Indicators. https://databank.worldbank.org/source/world-development-indicators#

World Bank (2024b), Digital Progress and Trends Report 2023, https://www.worldbank.org/en/publication/digital-progress-and-trends-report

World Bank (2023), Digital Progress and Trends Report 2023, https://openknowledge.worldbank.org/handle/10986/40970

World Bank (2020), DataBank: Doing Business (database) accessed February 2025, https://databank.worldbank.org/reports.aspx?source=3001&series=IC.ELC.SAID.XD.DB1619

WSJ (The Wall Street Journal) (2025), How AI Can Protect Vital Pipelines and Cables Deep in the Ocean, https://www.wsj.com/tech/ai/ai-military-applications-mapping-aca7f486

Annex A: Methodology and data tables

Forbes (2021), In 2020 HDD Companies Shipped Over 1ZB of Storage Capacity, https://www.forbes.com/sites/tomcoughlin/2021/02/07/in-2020-hdd-companies-shipped-over-1zb-of-storage-capacity/

Gartner (2020), Gartner Says Worldwide Server Revenue Grew 5.1% in the Fourth Quarter of 2019, While Shipments Increased 11.7%, https://www.gartner.com/en/newsroom/press-releases/2020-03-19-gartner-says-worldwide-server-revenue-grew-5-percent-in-the-fourth-quarter-of-2019-while-shipments-increased-11-percent

IEA. CC BY 4.0.

Gartner (2018a), Gartner Says Worldwide Server Revenue Grew 25.7 Percent in the Fourth Quarter of 2017, While Shipments Increased 8.8 Percent, https://www.gartner.com/en/newsroom/press-releases/2018-03-08-gartner-says-worldwide-server-revenue-grew-in-the-fourth-quarter-of-2017

Gartner (2018b), Gartner Says Worldwide Server Revenue Grew 33.4 Percent in the First Quarter of 2018, While Shipments Increased 17.3 Percent, https://www.gartner.com/en/newsroom/press-releases/2018-06-11-gartner-says-worldwide-server-revenue-grew-33-percent-in-the-first-quarter-of-2018-while-shipments-increased-17-percent

Gartner (2017a), Gartner Says Worldwide Server Revenue Grew 16 Percent in the Third Quarter of 2017; Shipments Grew 5.1 Percent, https://www.gartner.com/en/newsroom/press-releases/2017-12-11-gartner-says-worldwide-server-revenue-grew-16-percent-in-third-quarter-of-2017

Gartner (2017b), Gartner Says Worldwide Server Shipments Declined 4.2 Percent in the First Quarter of 2017; Revenue Declined 4.5 Percent, https://www.gartner.com/en/newsroom/press-releases/2017-06-07-gartner-says-worldwide-server-shipments-declined-4-percent-in-the-first-quarter-of-2017-revenue-declined-4-percent

Gartner (2017c), Gartner Says Worldwide Server Shipments Grew 2.4 Percent in the Second Quarter of 2017; Revenue Grew 2.8 Percent, https://www.gartner.com/en/newsroom/press-releases/2017-09-12-gartner-says-worldwide-server-shipments-grew-2-percent-in-the-second-quarter-of-2017-revenue-grew-2-percent

Gartner (2016a), Gartner Says Worldwide Server Revenue Declined 0.8 Percent in the Second Quarter of 2016, While Shipments Increased 2 Percent, https://www.gartner.com/en/newsroom/press-releases/2016-09-14-gartner-says-worldwide-server-revenue-declined-1-percent-in-the-second-quarter-of-2016-while-shipments-increased-2-percent

Gartner (2016b), Gartner Says Worldwide Server Revenue Grew 8.2 Percent in the Fourth Quarter of 2015, While Shipments Increased 9.2 Percent, https://www.gartner.com/en/newsroom/press-releases/2016-03-09-gartner-says-worldwide-server-revenue-grew-8-percent-in-the-fourth-quarter-of-2015-while-shipments-increased-9-percent

Gartner (2015a), Gartner Says Worldwide Server Market Grew 4.8 Percent in Shipments, While Revenue Increased 2.2 Percent in Fourth Quarter of 2014, https://www.gartner.com/en/newsroom/press-releases/2015-03-03-gartner-says-worldwide-server-market-grew-4-percent-in-shipments-while-revenue-increased-2-percent-in-fourth-quarter-of-2014

Gartner, (2015b), Gartner Says Worldwide Server Revenue Grew 7.5 Percent in the Third Quarter of 2015, While Shipments Increased 9.2 Percent, https://www.gartner.com/en/newsroom/press-releases/2015-12-02-gartner-says-worldwide-server-revenue-grew-7-percent-in-the-third-quarter-of-2015-while-shipments-increased-9-percent

IEA. CC BY 4.0.

Gartner (2015c), Gartner Says Worldwide Server Shipment Market Grew 8 Percent in the Second Quarter of 2015, While Revenue Increased 7.2 Percent, https://www.gartner.com/en/newsroom/press-releases/2015-08-26-gartner-says-worldwide-server-shipment-market-grew-8-percent-in-the-second-quarter-of-2015-while-revenue-increased-7-percent

Gartner (2015d), Gartner Says Worldwide Server Shipments Grew 13 Percent in the First Quarter of 2015, While Revenue Increased 17.9 Percent, https://www.gartner.com/en/newsroom/press-releases/2015-05-28-gartner-says-worldwide-server-shipments-grew-13-percent-in-the-first-quarter-of-2015-while-revenue-increased-18-percent

Gartner (2014a), Gartner Says Worldwide Server Shipments Grew 1 Percent in the Third Quarter of 2014 While Revenue Increased 1.7 Percent, https://www.gartner.com/en/newsroom/press-releases/2014-12-03-gartner-says-worldwide-server-shipments-grew-1-percent-in-the-third-quarter-of-2014-while-revenue-increased-1-percent

Gartner (2014b), Gartner Says Worldwide Server Shipments Market Grew 1.3 Percent in the Second Quarter of 2014 While Revenue Increased 2.8 Percent, https://www.gartner.com/en/newsroom/press-releases/2014-08-27-gartner-says-worldwide-server-shipments-market-grew-1-percent-in-the-second-quarter-of-2014-while-revenue-increased-2-percent

Gartner, (2014c), Gartner Says Worldwide Server Shipments Market Grew 1.4 Percent in the First Quarter of 2014, While Revenue Declined 4.1 Percent, https://www.gartner.com/en/newsroom/press-releases/2014-05-28-gartner-says-worldwide-server-shipments-market-grew-1-percent-in-the-first-quarter-of-2014-while-revenue-declined-4-percent

Google (2025), Google DataCenters: Efficiency, https://datacenters.google/efficiency/

Hintemann, R., Hinterholtzer, S. and Konrat, F. (2024), Server Stock Data — A Basis for Determining the Energy and Resource Requirements of Data Centres, 2024 Electronics Goes Green 2024+ (EGG), pp. 1-5, https://ieeexplore.ieee.org/abstract/document/10631194

IDC (2024), Datacenter Trends: Sustainable Builds and Carbon Emissions, https://www.idc.com/getdoc.jsp?containerId=IDC_P33186

Koomey, J. (2011), Growth in data center electricity use 2005 to 2010. A report by Analytical Press, completed at the request of The New York Times, https://alejandrobarros.com/wp-content/uploads/old/Growth_in_Data_Center_Electricity_use_2005_to_2010.pdf

Koomey, J. G. (2007), Estimating total power consumption by servers in the U.S. and the world, http://uploadi.www.ris.org/editor/1203418697svrpwrusecompletefinal.pdf

Lei, N. and Masanet, E. (2022), Climate- and technology-specific PUE and WUE estimations for U.S. data centers using a hybrid statistical and thermodynamics-based approach, https://www.sciencedirect.com/science/article/pii/S0921344922001719

Malmodin, J. et al. (2024), ICT sector electricity consumption and greenhouse gas emissions – 2020 outcome, https://www.sciencedirect.com/science/article/pii/S0308596123002124

IEA. CC BY 4.0.

Masanet, E. et al. (2020), Recalibrating global data center energy-use estimates. Science, vol. 367(6481), pp. 984-986, https://www.science.org/doi/abs/10.1126/science.aba3758

OMDIA (2025), Data Center Building and Investment Intelligence Service, https://omdia.tech.informa.com/advance-your-business/cloud-and-data-center/data-center-building-and-investment-intelligence-service

SemiAnalysis (2025), Custom Market Intelligence, https://semianalysis.com/datacenter-industry-model/

Shehabi, A., et al. (2024), 2024 United States Data Center Energy Usage Report, https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf

Shehabi, A., et al. (2018), Data center growth in the United States: decoupling the demand for services from electricity use, https://iopscience.iop.org/article/10.1088/1748-9326/aaec9c

Shehabi, A., et al. (2016), United States Data Center Energy Usage Report, https://eta.lbl.gov/publications/united-states-data-center-energy

SPEC (2024), SPECpower_ssj 2008, https://www.spec.org/power_ssj2008/results/

Statista (2024), Data Center – Worldwide, https://www.statista.com/outlook/tmo/data-center/worldwide

Turner & Townsend (2024), Data centre cost index 2024, https://reports.turnerandtownsend.com/dcci-2024/

World Bank (2016), World Development Report 2016: Digital Dividends, https://www.worldbank.org/en/publication/wdr2016

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